

Day 3

The Internet of Things Stack

My classes (4)

~~● Business models and the Internet of Things.~~

~~● Business Model Canvas~~

~~—~~
~~● IoT / Edge AI project (hands on)~~

~~—~~
● The IoT Stack - How to build your IoT project, digital transformation and more.

● IoT Model Canvas

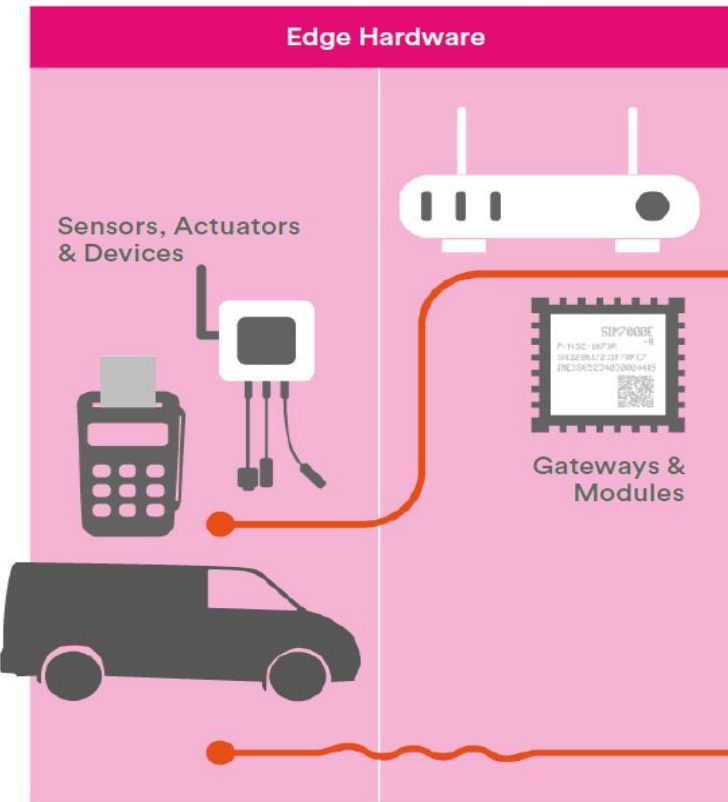
~~—~~
● Scale your IoT / Edge AI projects (hands-on)

Today

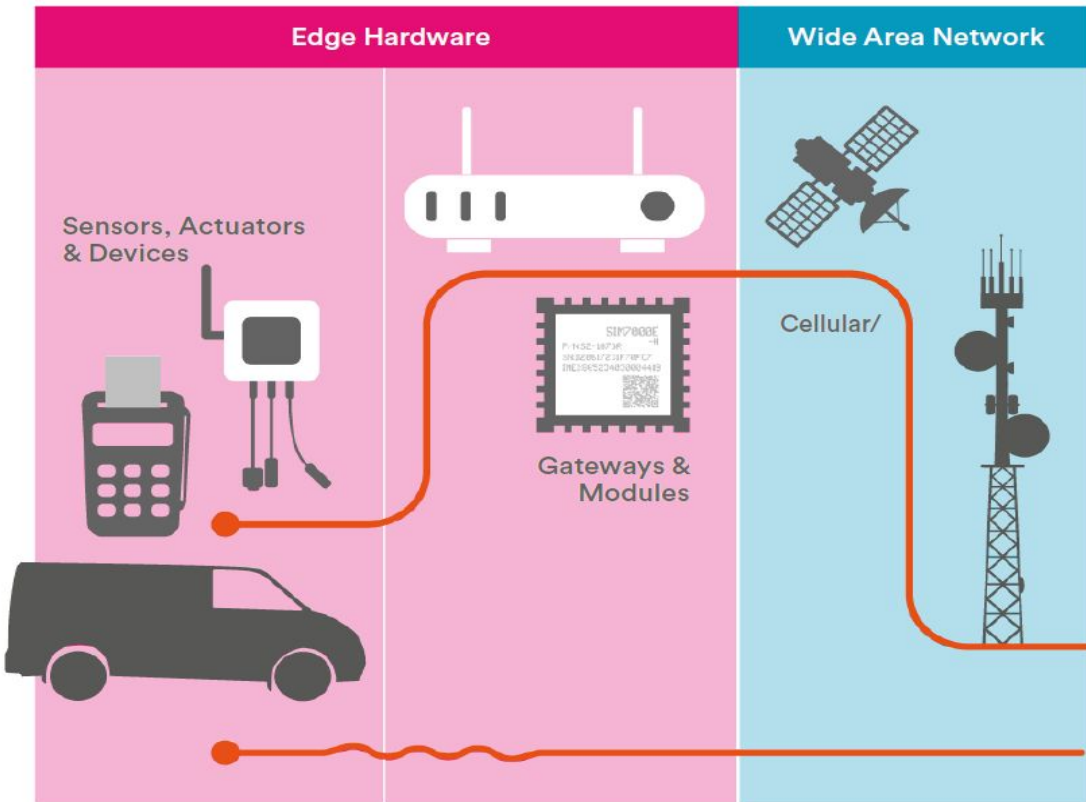
- What do you need to build an IoT Solution?
- The IoT Stack
- The EU CRA (Cyber Resilience Act)
- IoT Model Canvas
- The Industrial IoT Use Case

What do you need to build an IoT Solution?

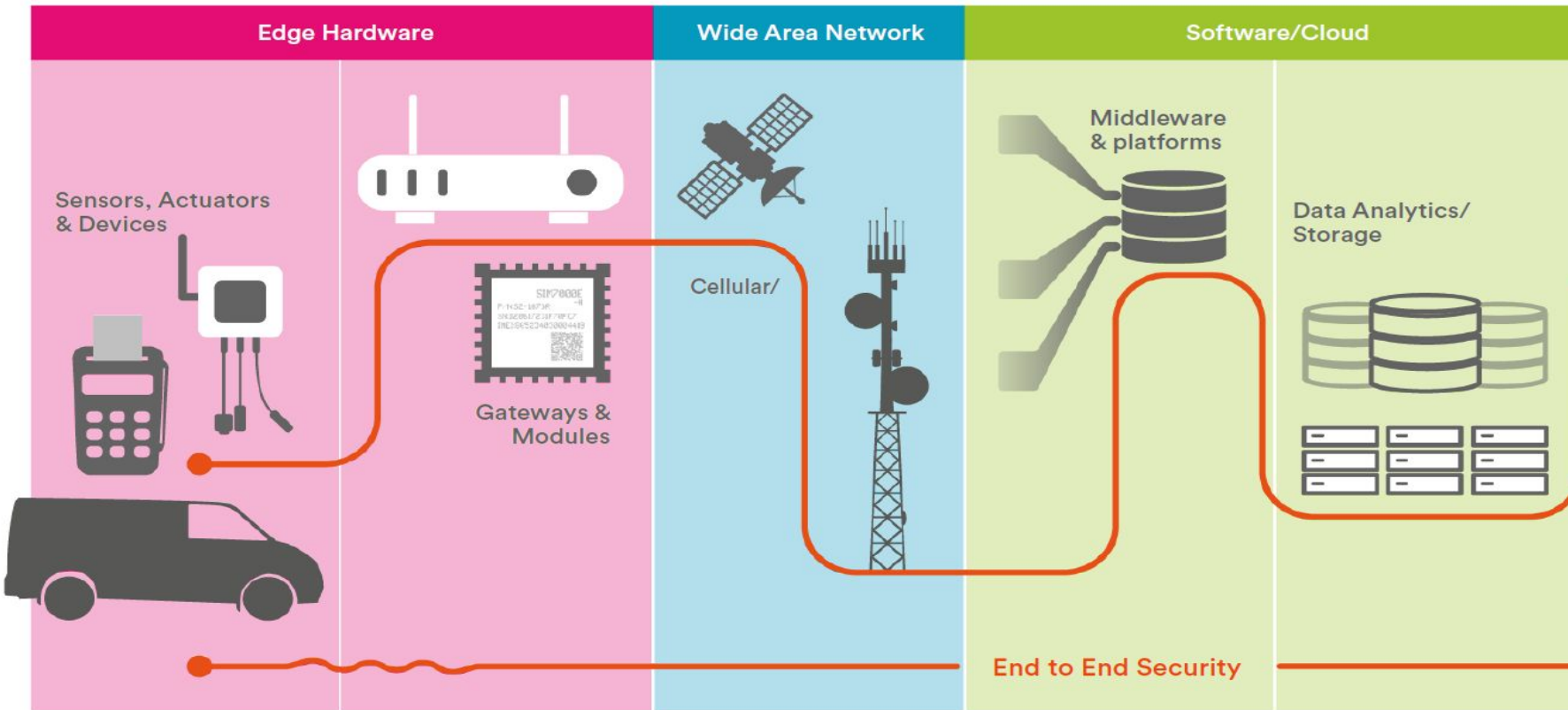
What do you need to build an IoT Solution?



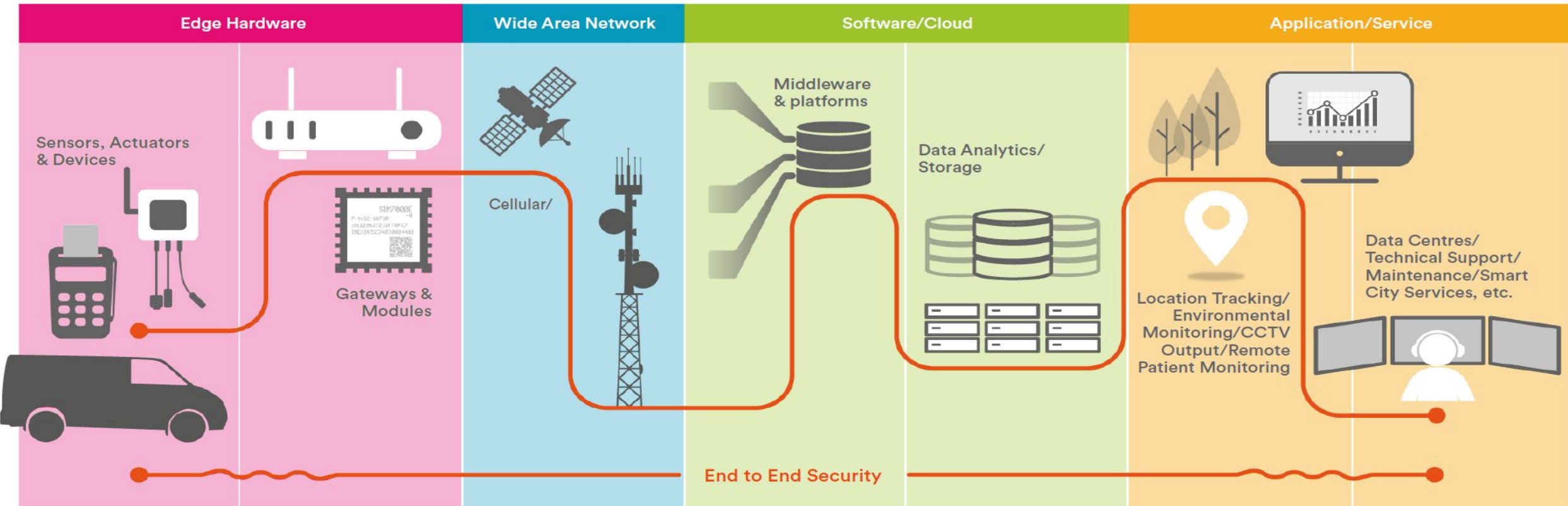
What do you need to build an IoT Solution?



What do you need to build an IoT Solution?

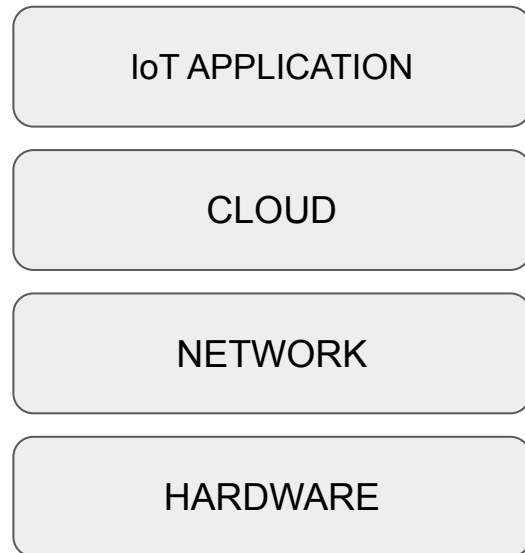


What do you need to build an IoT Solution?

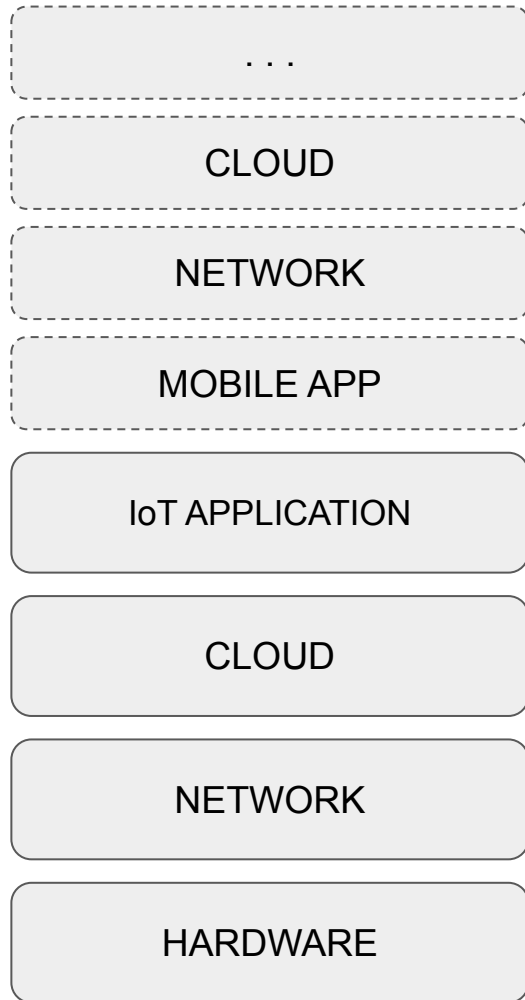


The IoT Stack

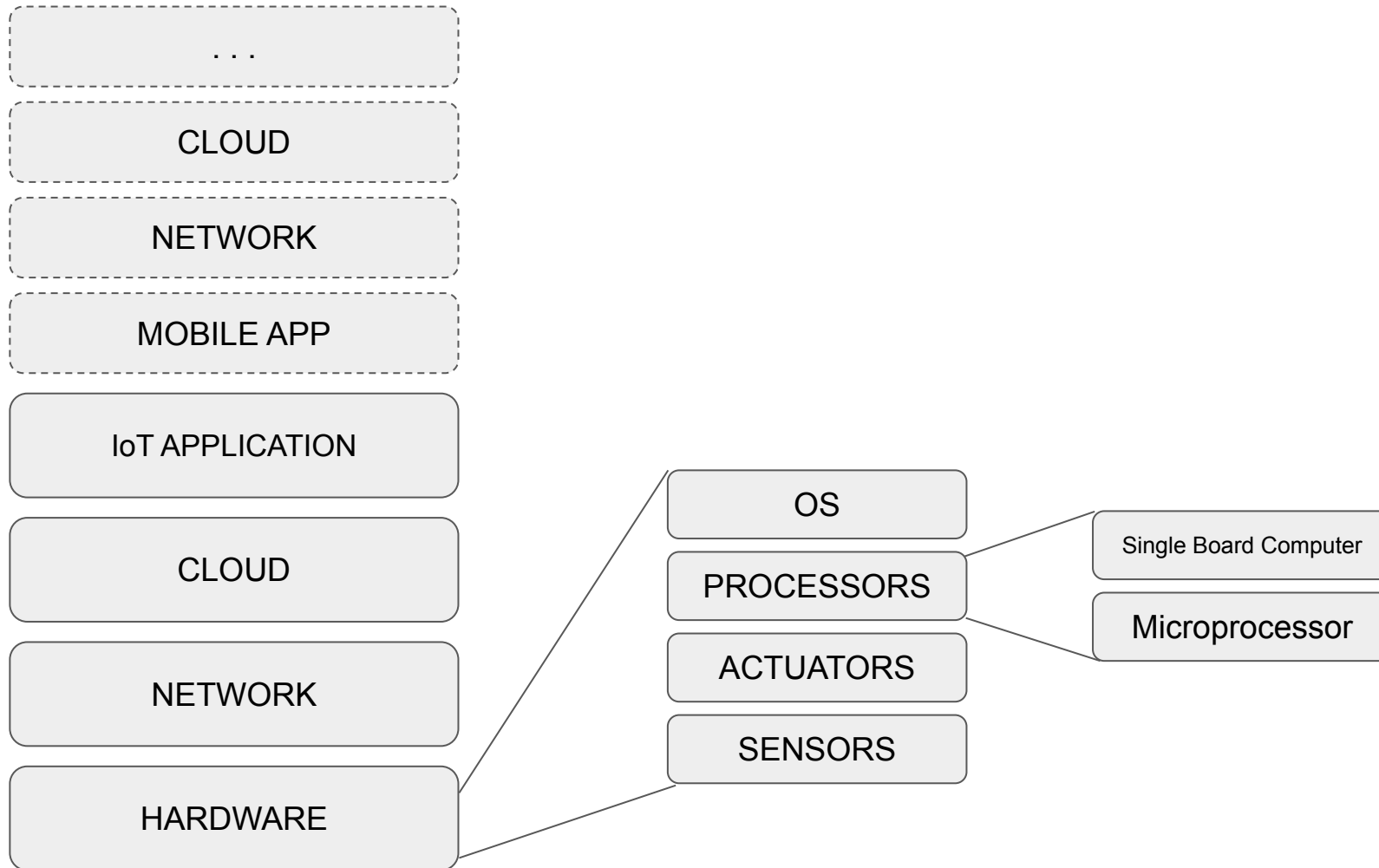
The IoT Stack



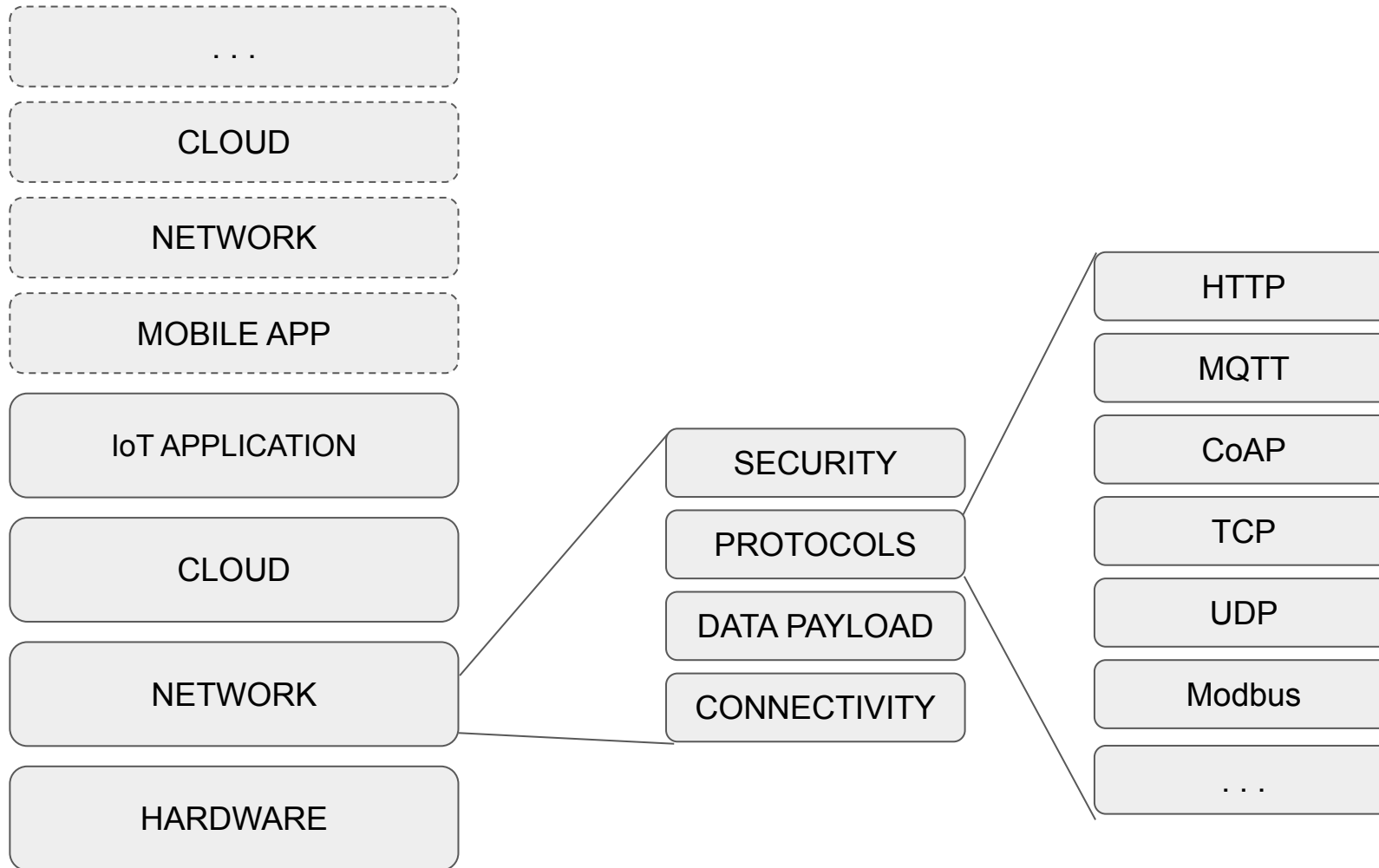
The IoT Stack



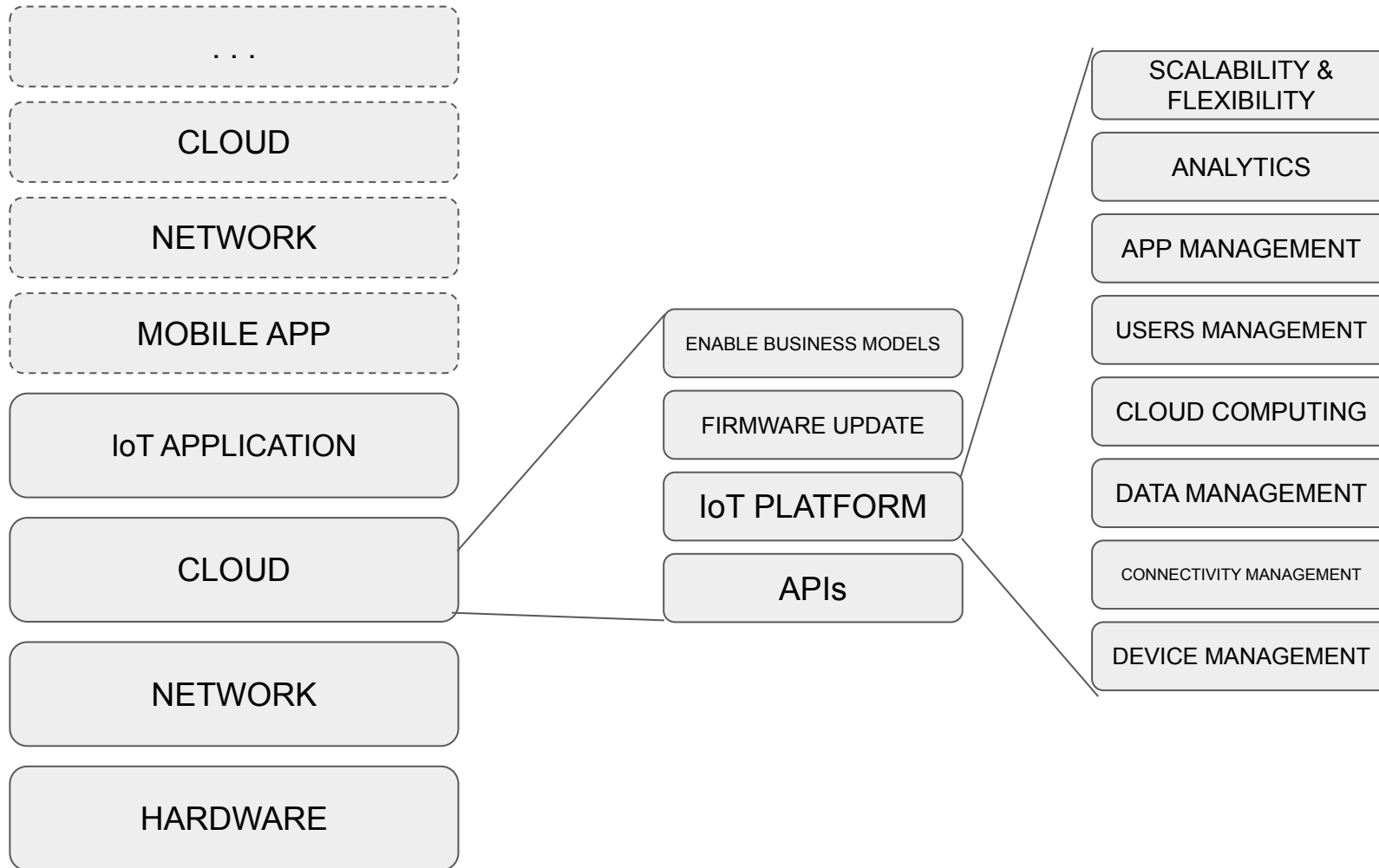
The IoT Stack



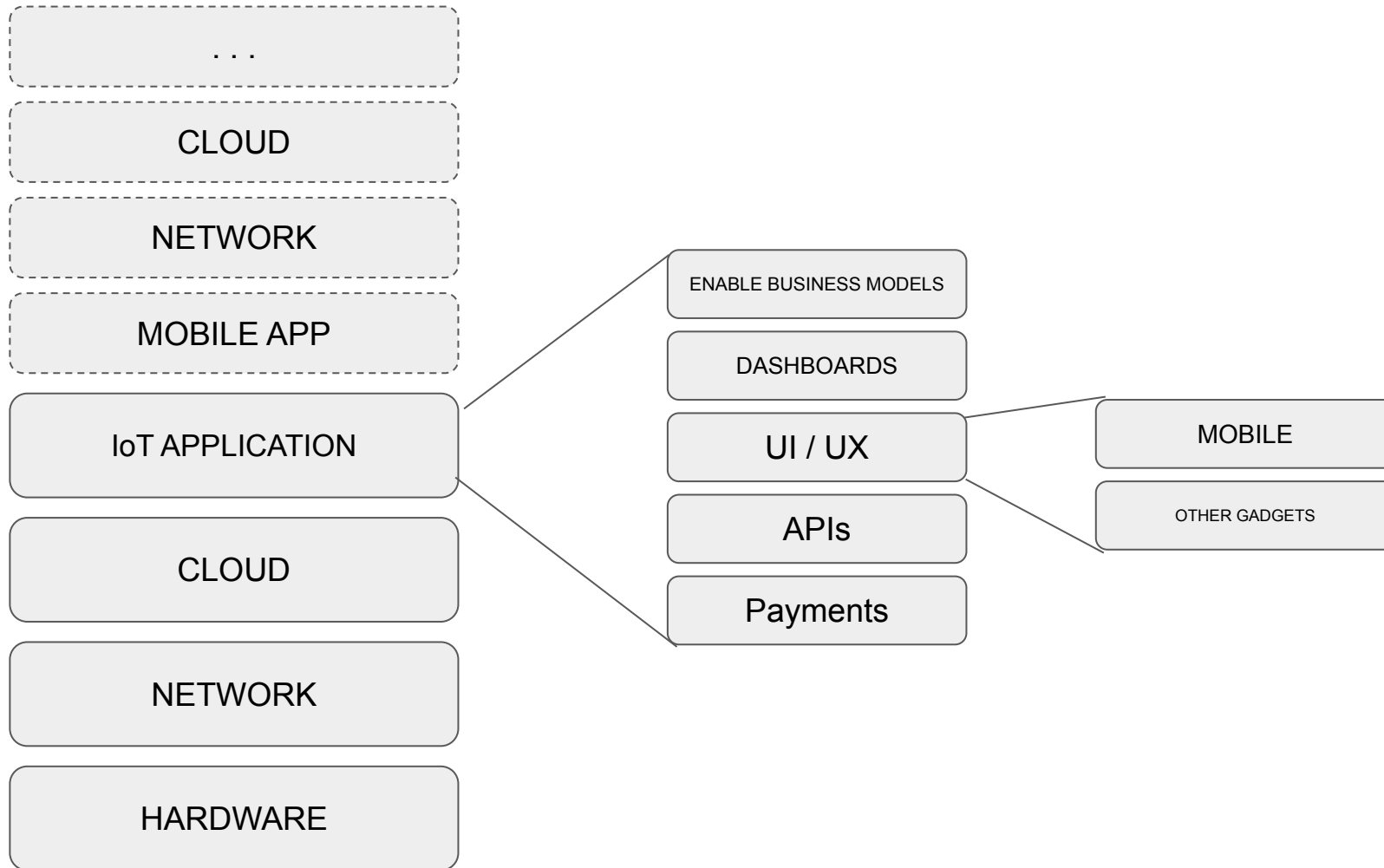
The IoT Stack



The IoT Stack



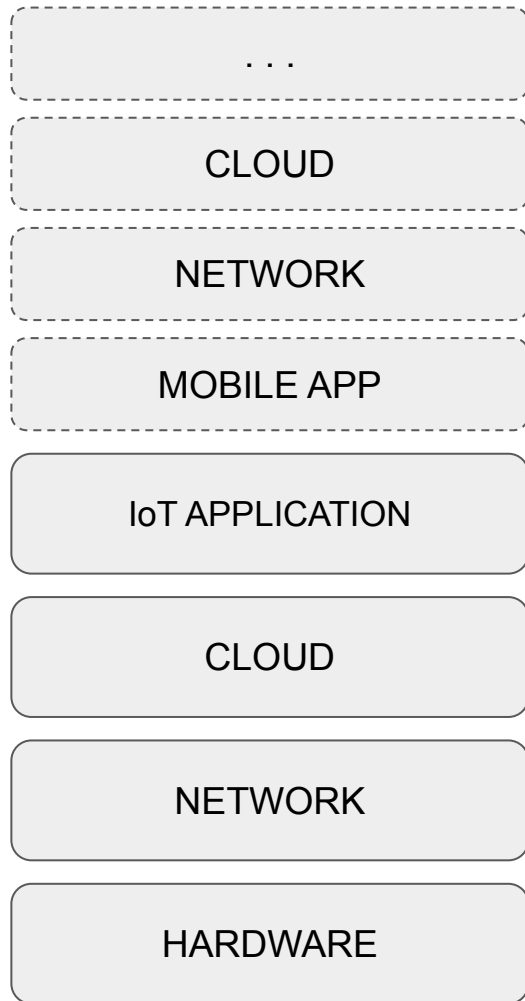
The IoT Stack



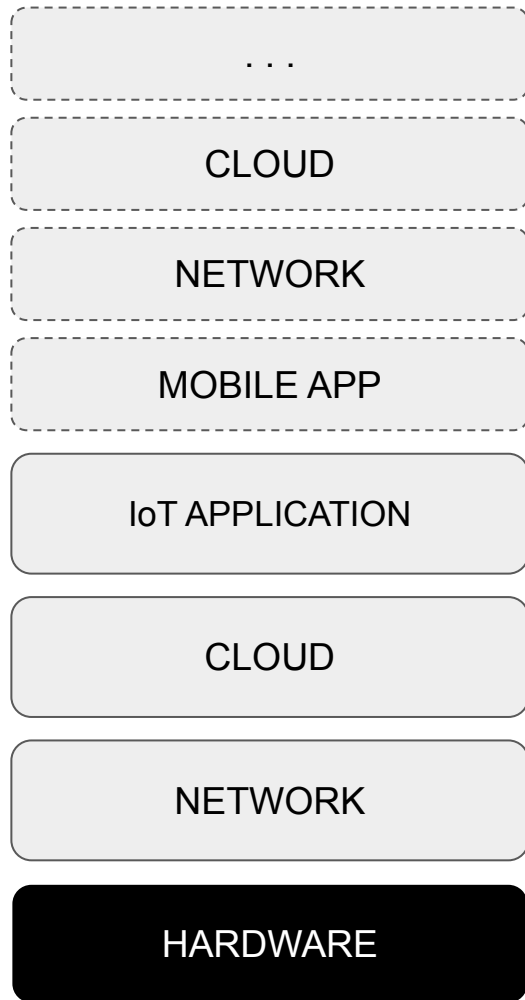
But still remember about the IoT definition...



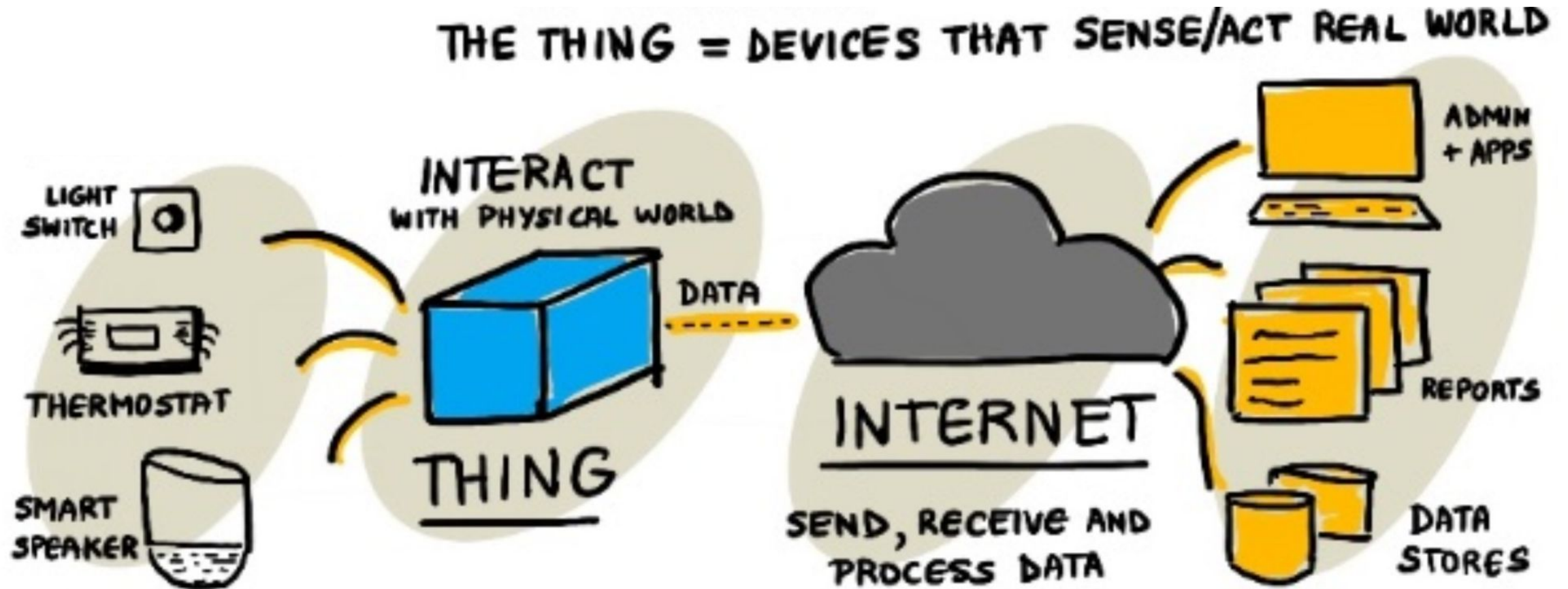
The IoT Stack (Zoom In)



The IoT Stack (Zoom In)



The hardware



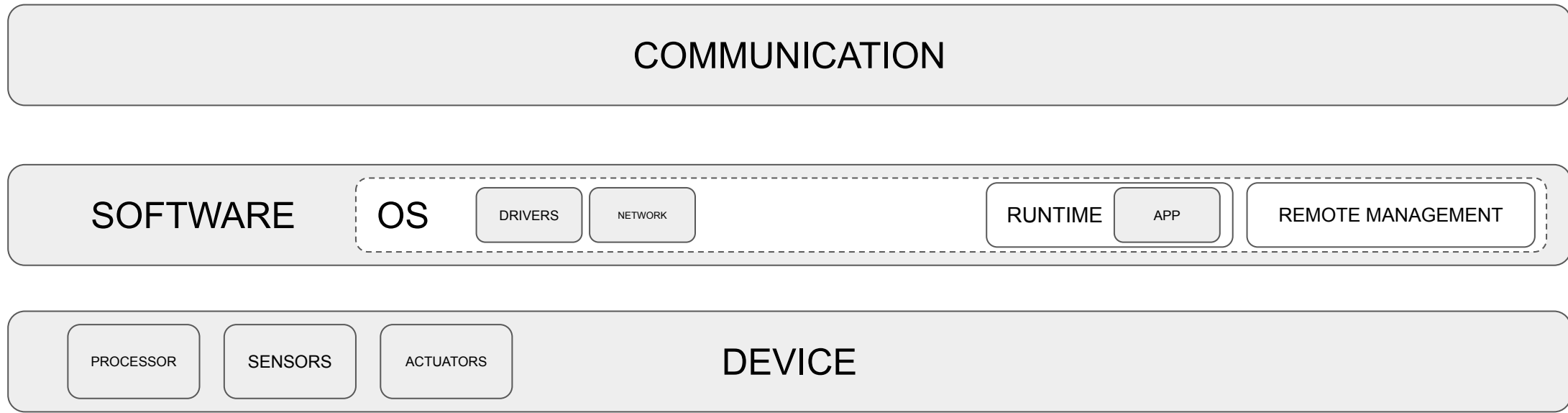
The hardware

COMMUNICATION

SOFTWARE

DEVICE

The hardware



Processors

Processors

- How much processing power do you need?
- How much storage do you need?
- How many sensors do you need to read?
- Do you need real-time control?
- Does your application run in the edge? (e.g. perform analytics on the edge, Machine Learning, etc.)
- Do you have enough processing power to support future software upgrade / releases?
- Do you have space constraints?
- Are we running on batteries?
- Do you need to perform AI (Floating point operations?)

Processors

MICROCONTROLLER

A microcontroller is a small computer with low-memory and programmable input / output peripherals.

- Can't run a complete operating system (e.g. freertos)
- Don't have the amount of computing power and resources.
- Run just one program repeatedly.

Examples

- <https://www.arduino.cc/>
- <https://www.espressif.com/en/products/socs/esp32>
- <https://www.st.com/en/microcontrollers-microprocessors/stm32-32-bit-arm-cortex-mcus.html>

SINGLE BOARD COMPUTER

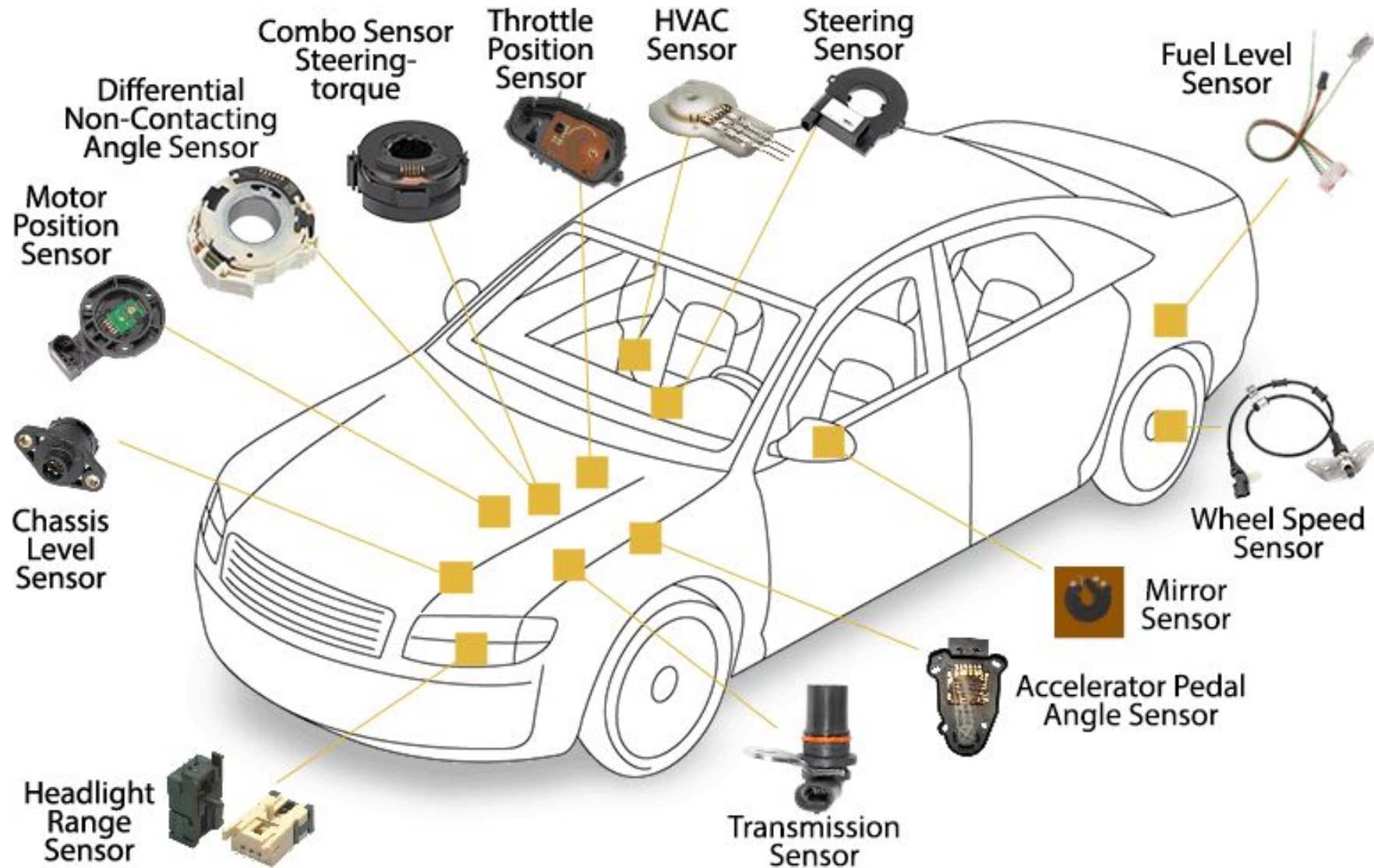
A Single Board Computer is a computer built on a single circuit board.

- Run only with operating systems.
- Can run more than one application at the same time.
- Need power to run all of the extra software.

Examples

- [Raspberry Pi](#)
- [Intel NUC](#)
- [NVIDIA Jetson](#)

The Sensors and Actuators



The Operating System (OS)

MICROCONTROLLER



Apache Nuttx



mbed OS

SINGLE BOARD COMPUTER



debian

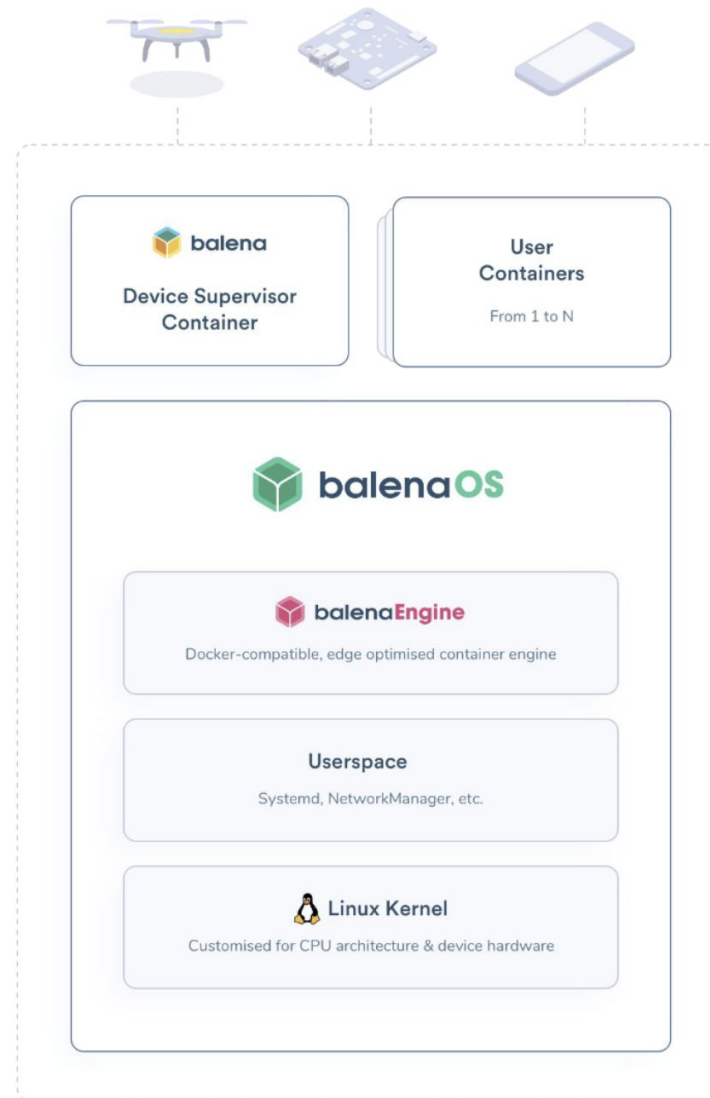


ubuntu



OTA Firmware update process: The balena case

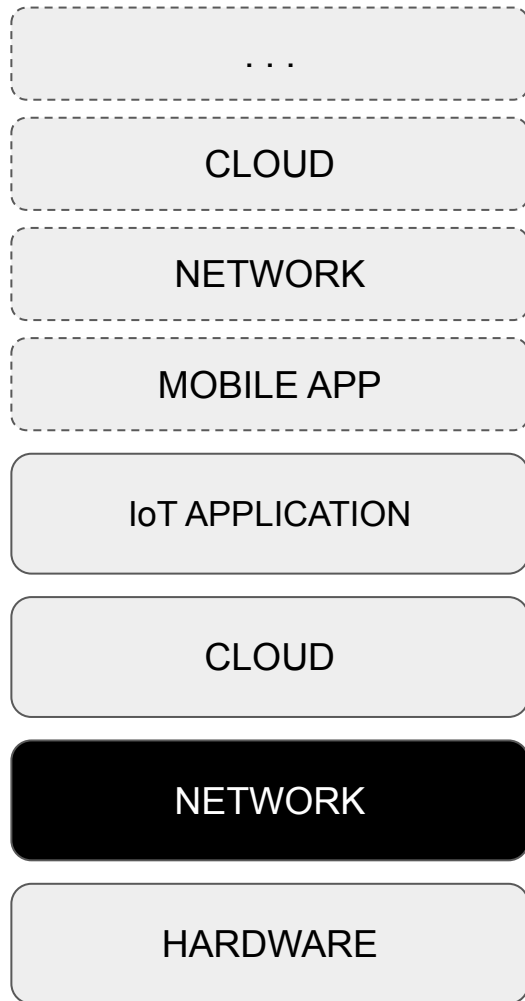
What is balena?



How it works



The IoT Stack (Zoom In)



Connectivity

- How are the devices connected to the Internet? (e.g. cellular communication, Bluetooth, WiFi, LoRa, Ethernet?)
- When devices are going to send data? (e.g. real-time, once a day, when an event happens?)
- Which kind of data is collected? (e.g. text, images, video?)
- Do they need bidirectional communication? or pulling once a day is enough?
- Does the communication module support secure encryption? What are the constraints of the hardware used?
- What QoS (Quality of Service) do you need on your communications? (e.g. is it ok to lose some messages or every message is critical)

Connectivity, Protocols and Serialization



NETWORKS



http://



CoAP



PROTOCOLS



MessagePack

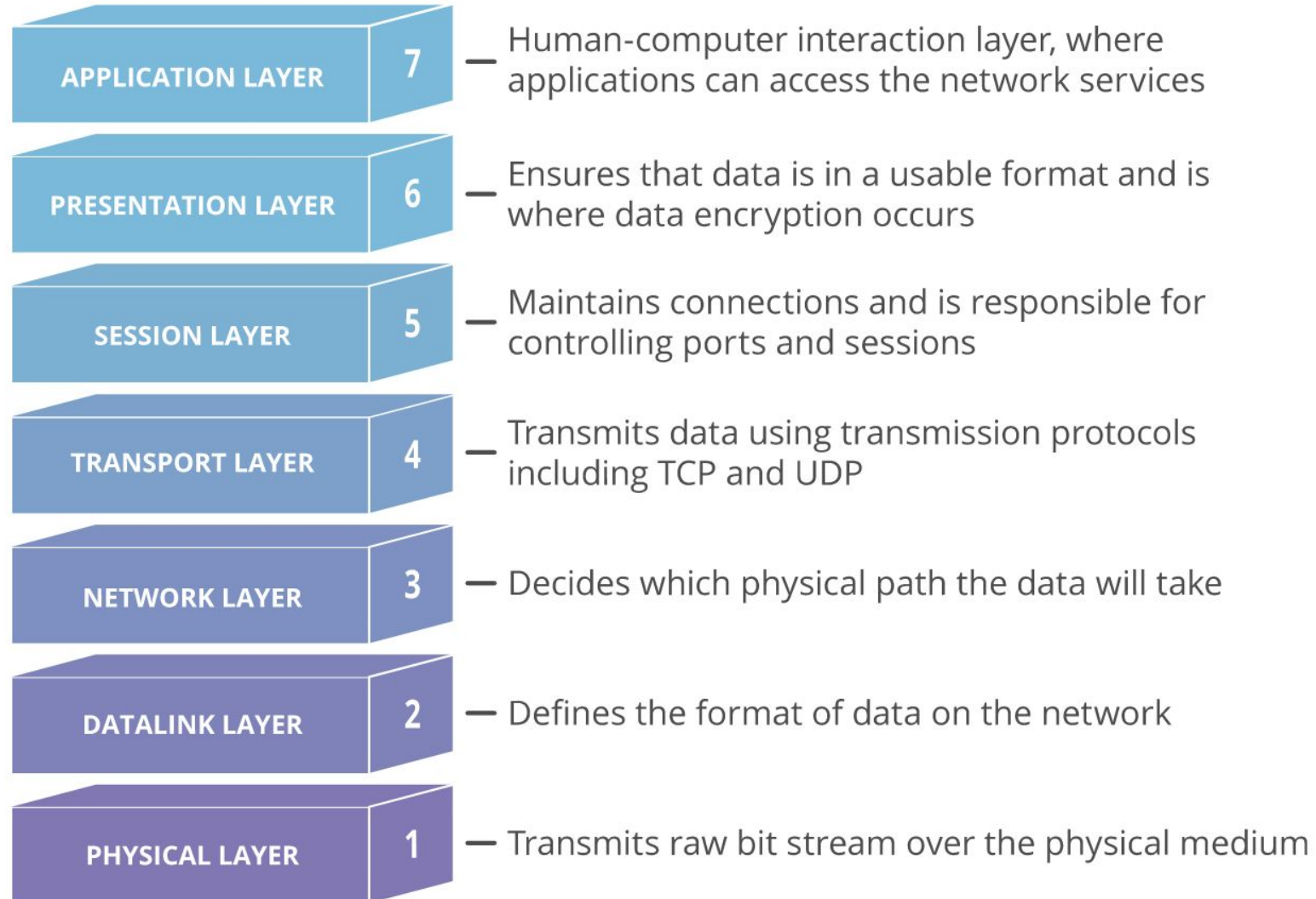


SERIALIZATION

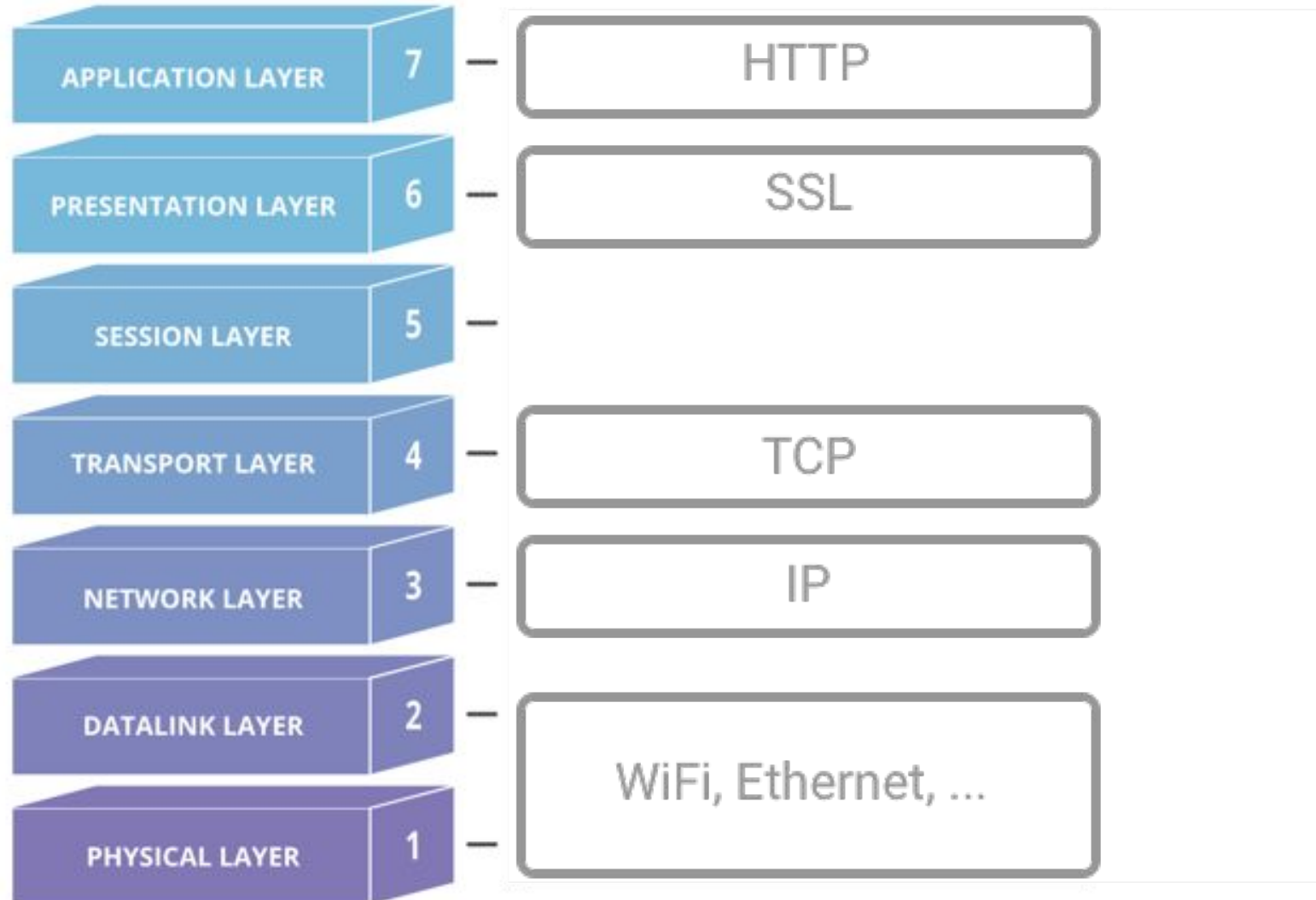


What is the best protocol to send data?

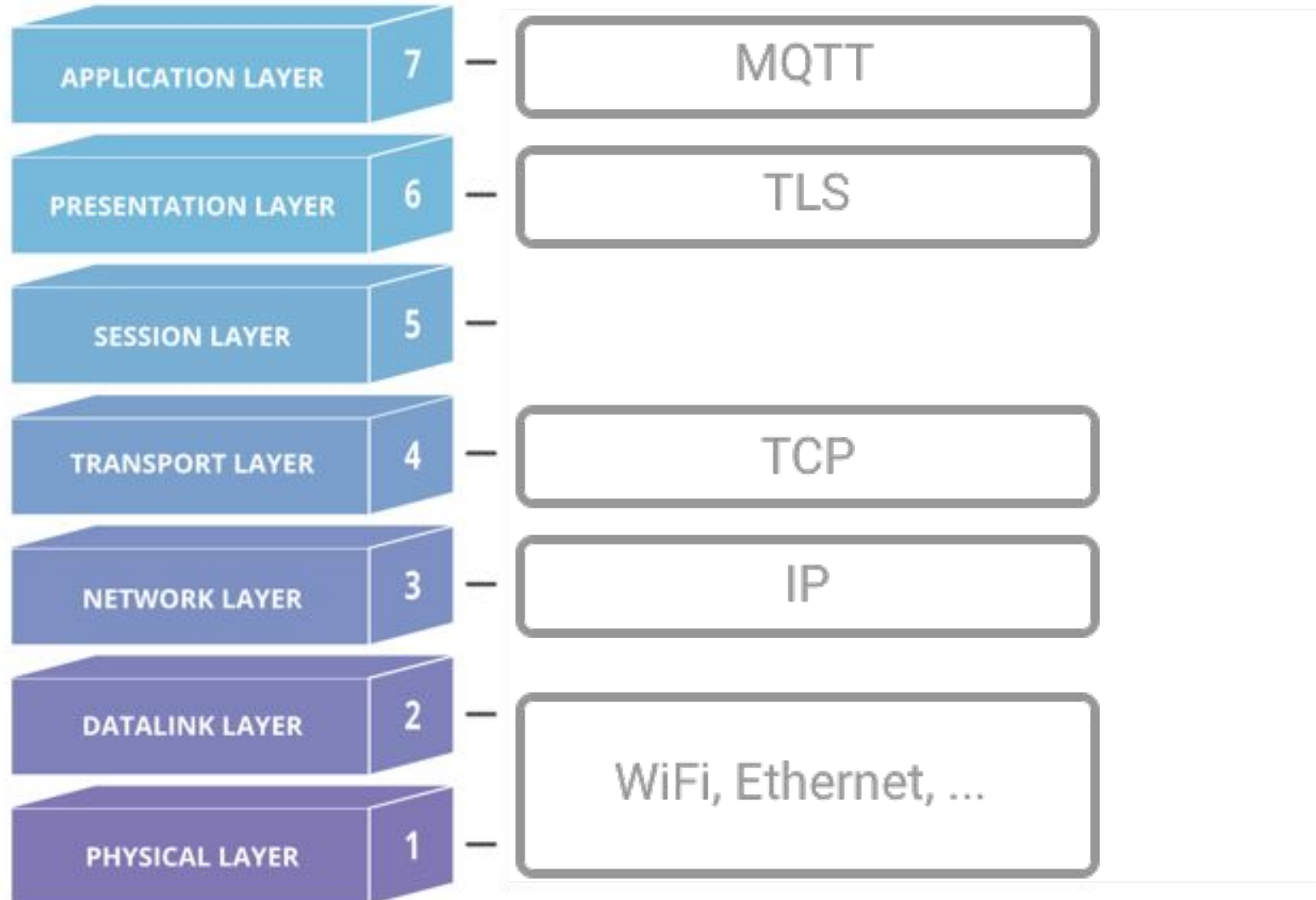
The OSI model



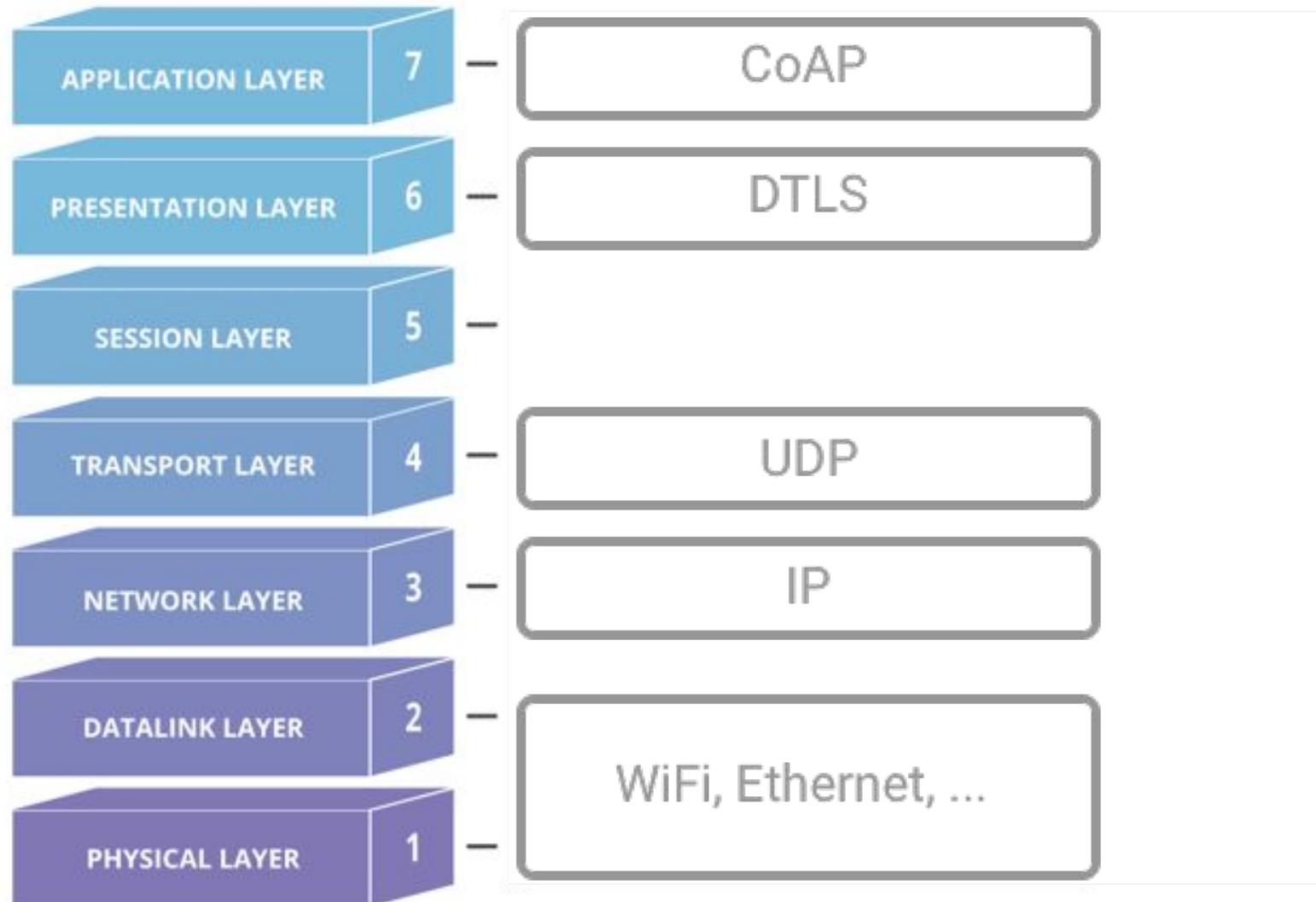
HTTP



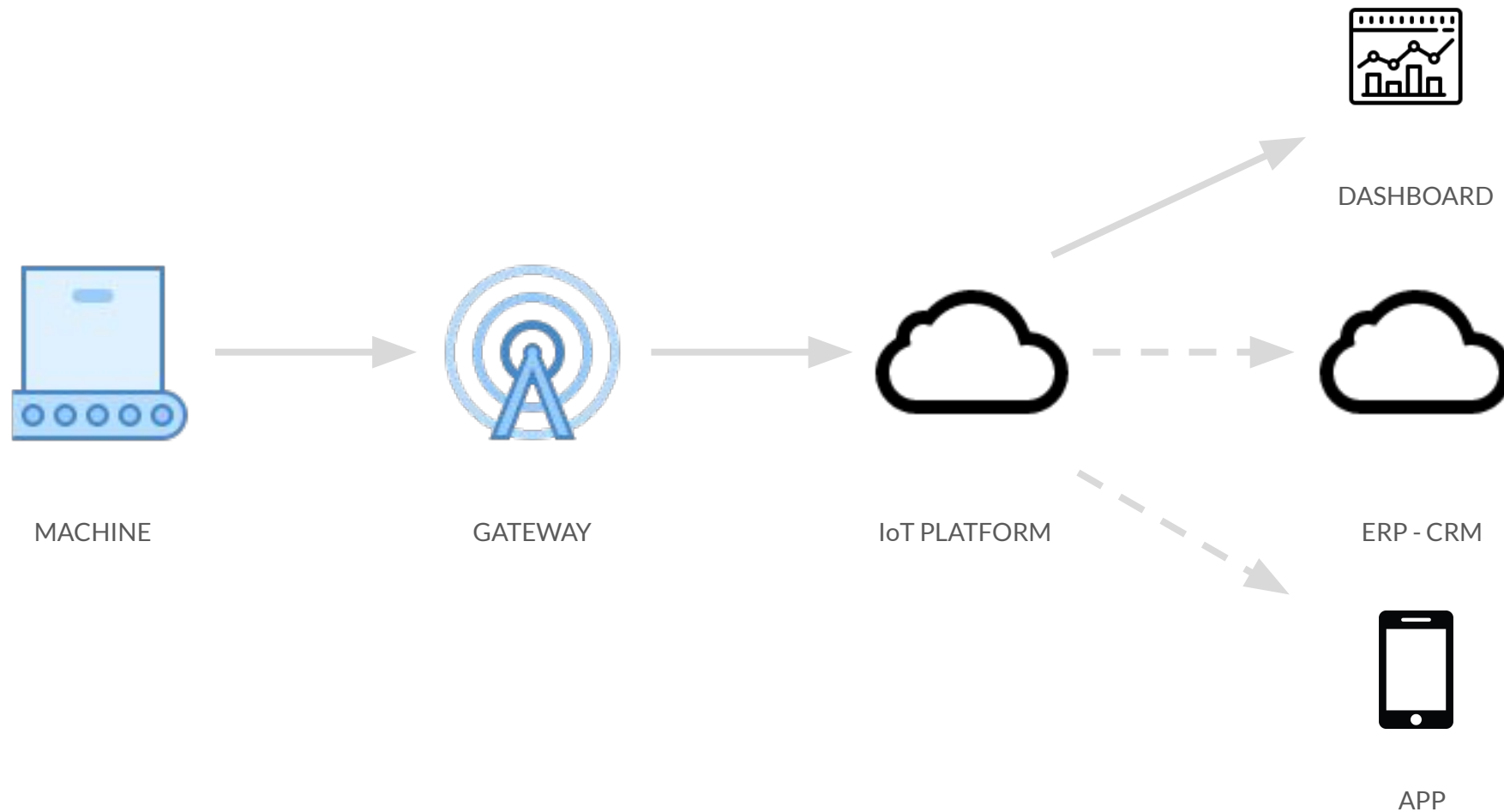
MQTT



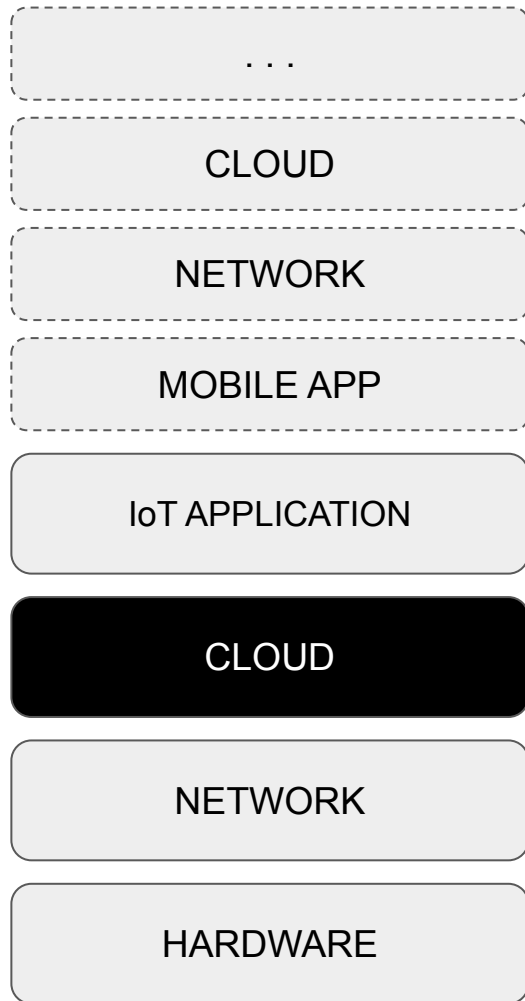
CoAP



Connectivity via Gateways



The IoT Stack (Zoom In)



The Cloud

The cloud is usually the backbone of your IoT solution.

The Cloud

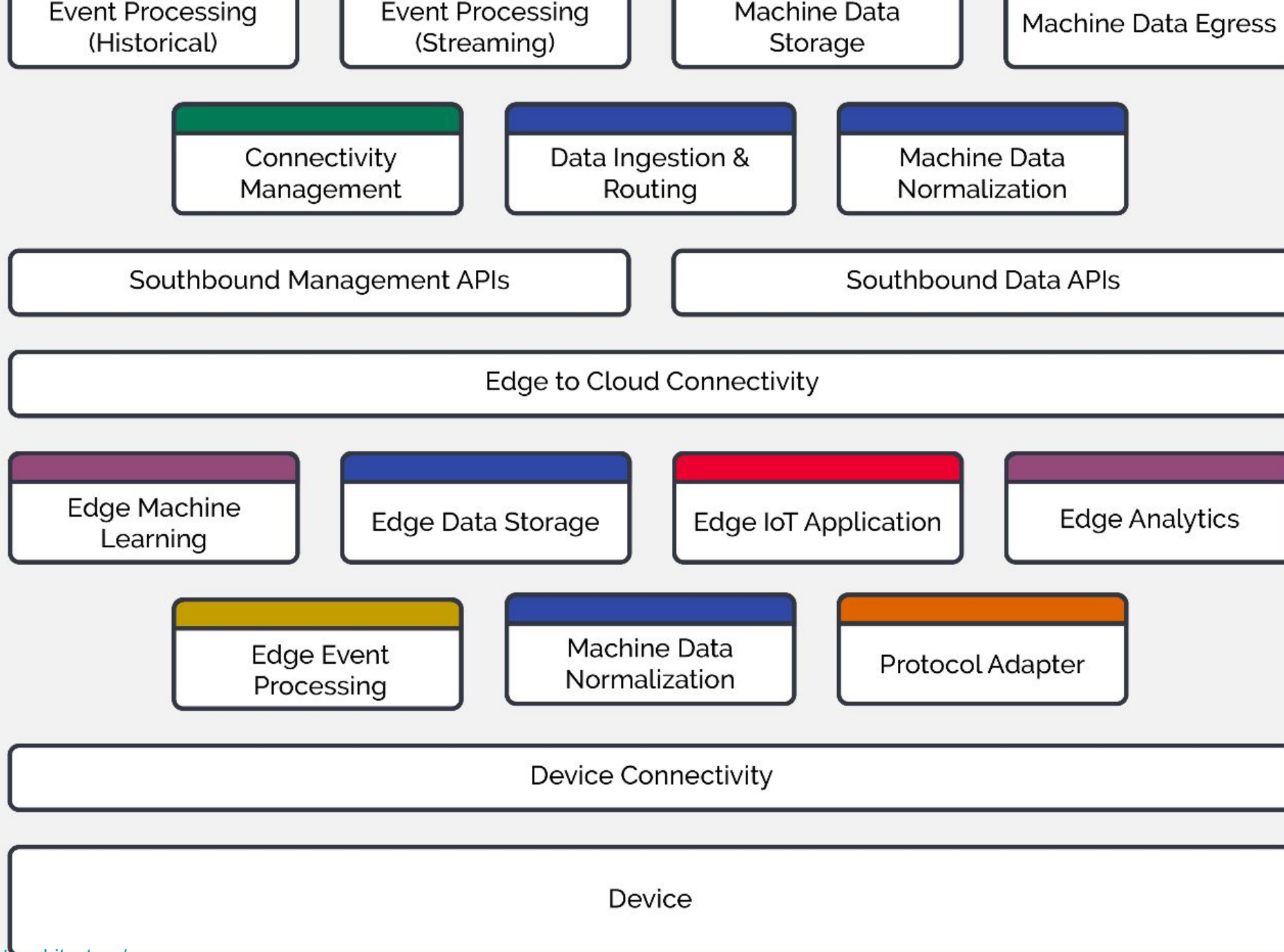
The cloud is usually the backbone of your IoT solution.

An IoT cloud platform have to support some critical areas:

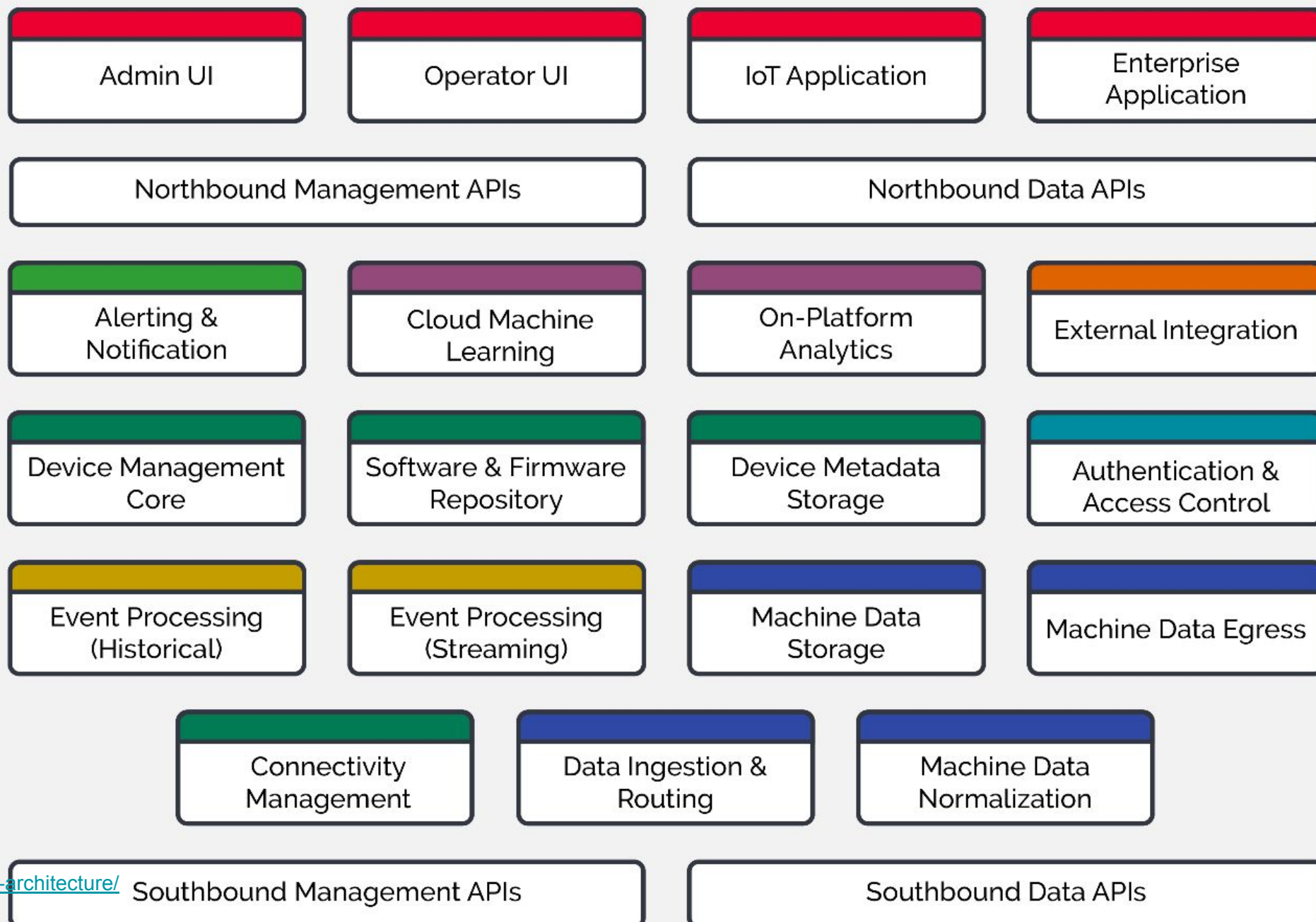
- Analytics and monitoring
- Devices
- Connectivity
- Data
- Users
- Apps
- APIs
- Cloud computing
- Scalability and flexibility
- Enable a business model

EDGE

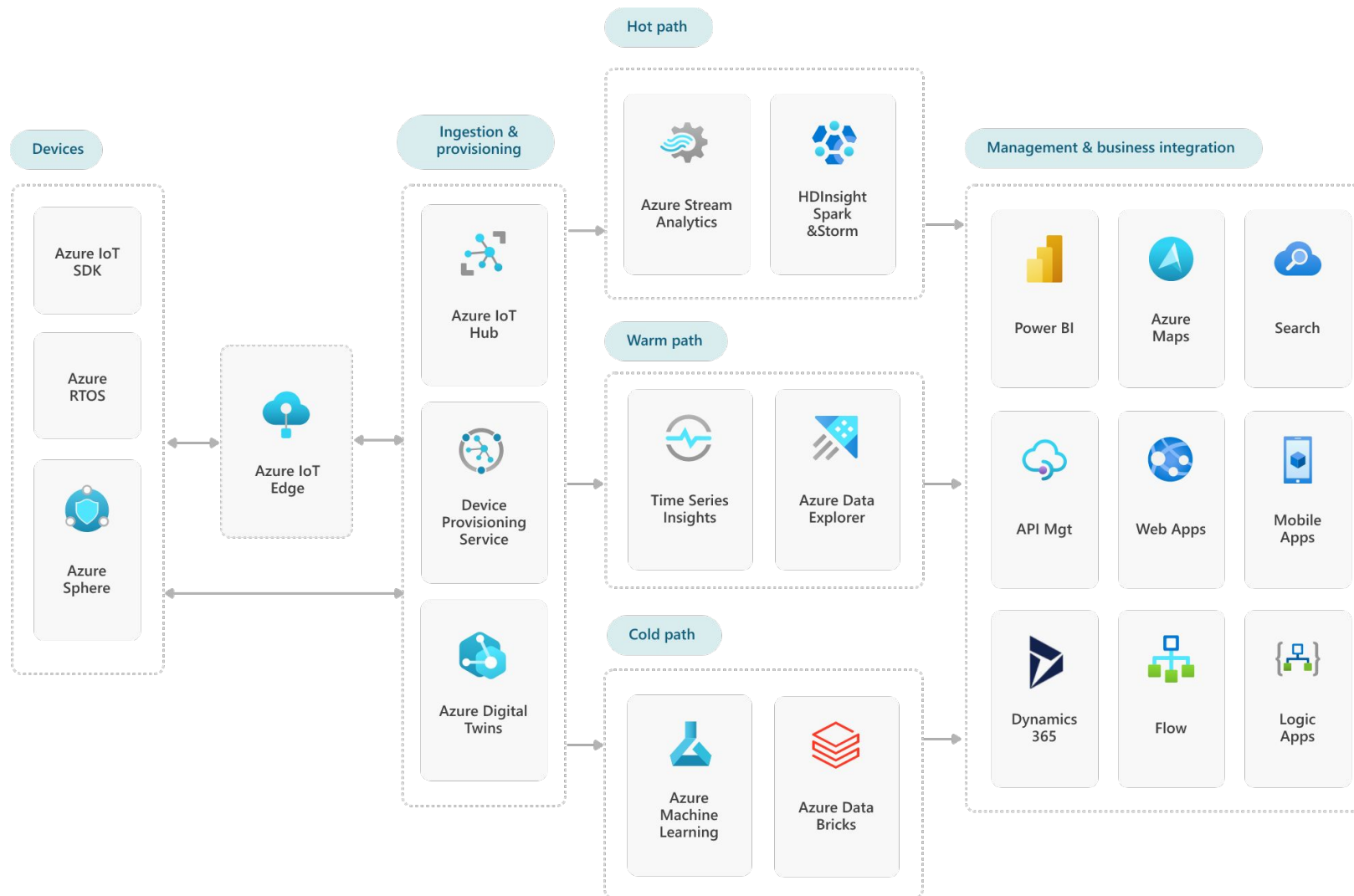
DEVICE



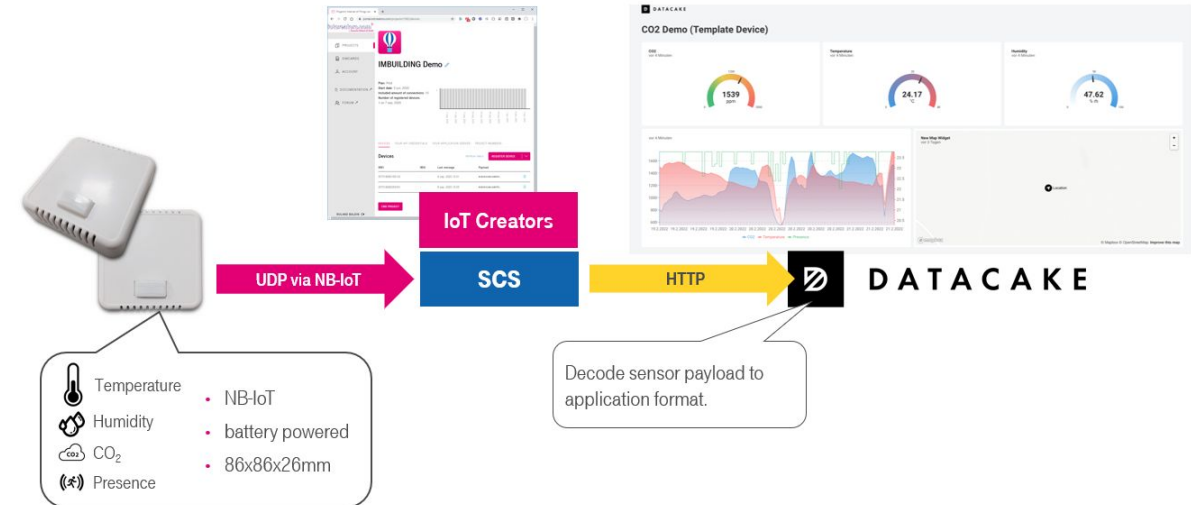
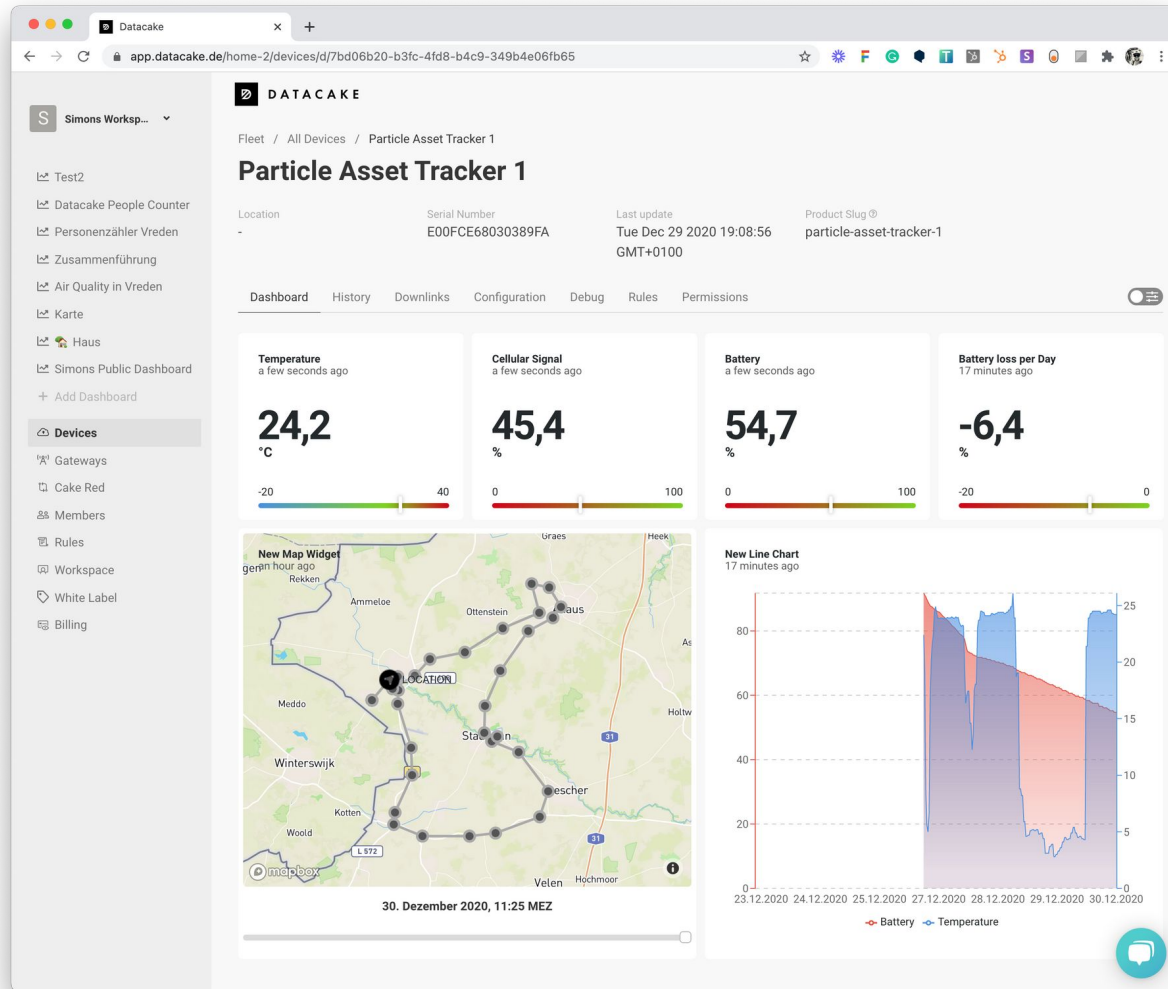
CLOUD



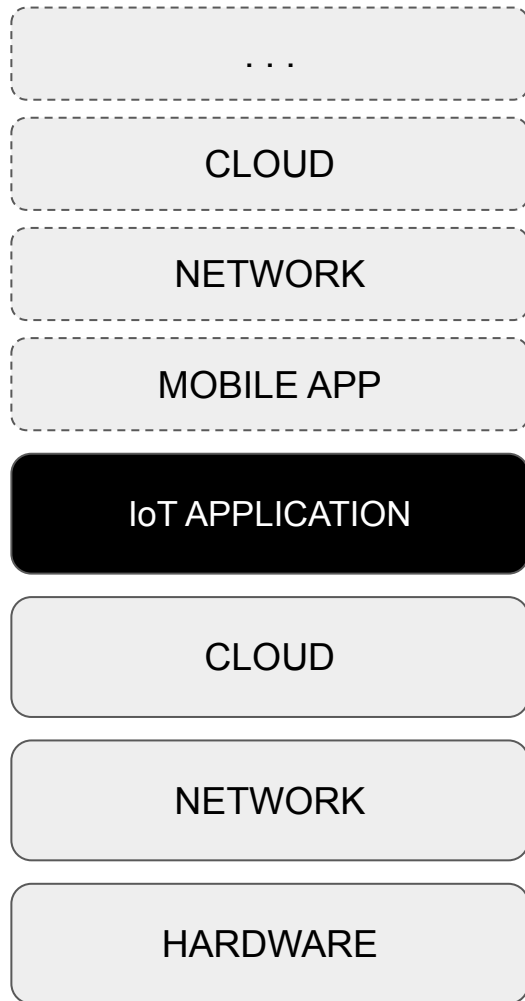
Example: The IoT Stack by Microsoft Azure



Example: Datacake



The IoT Stack (Zoom In)



The APIs

API (Application Programming Interface).

APIs are the key of a successful project.

If you want to integrate your IoT product with a 3rd party you need an API (and other protocols compatible).

Remember the southbound and northbound APIs from the architecture sketch.

The Applications

The Applications are based on the APIs.

Your resulting application depends on your use case and your business model.

What's your application?

The Applications

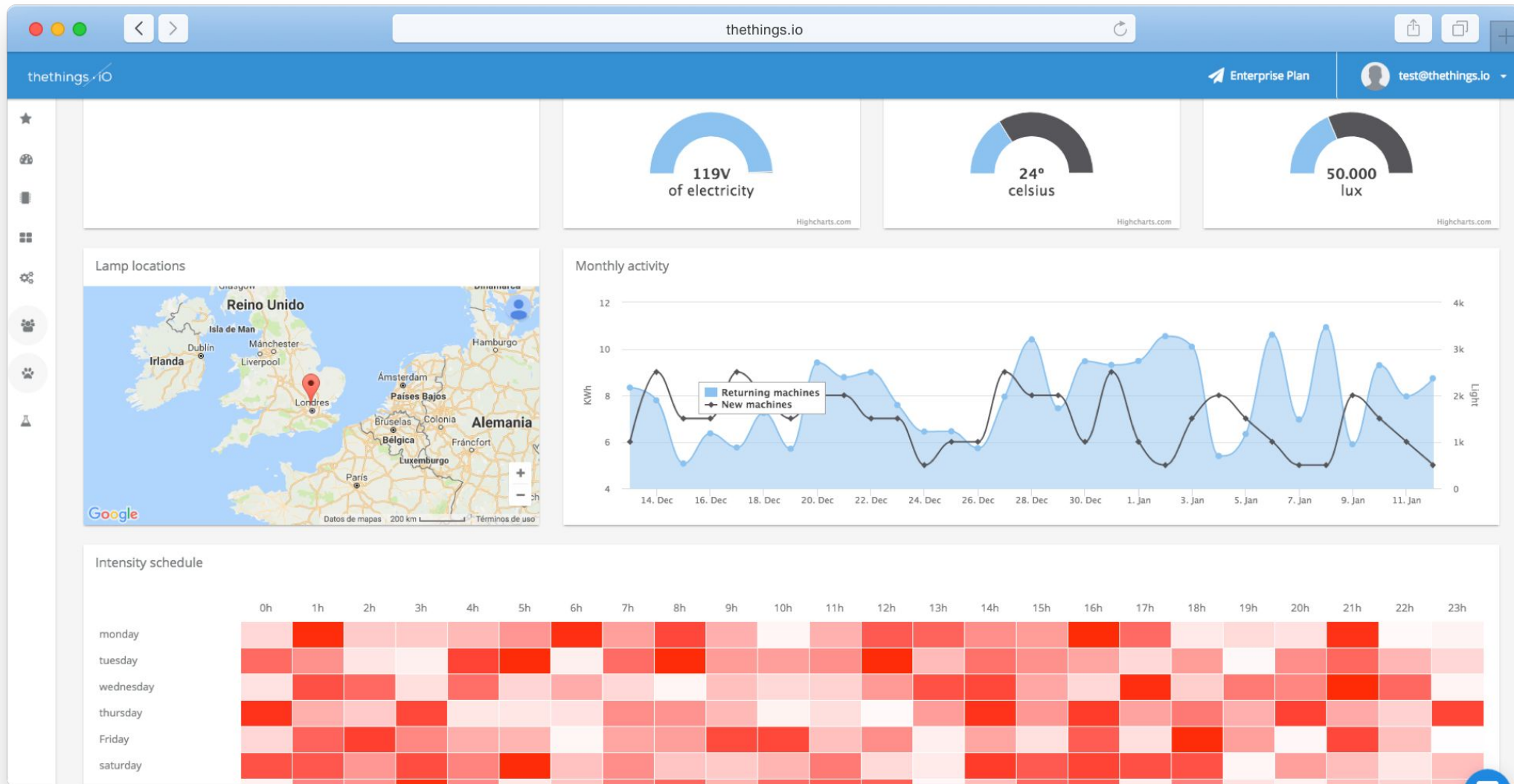
Is the mobile device the right tool for your IoT solution?

The Applications

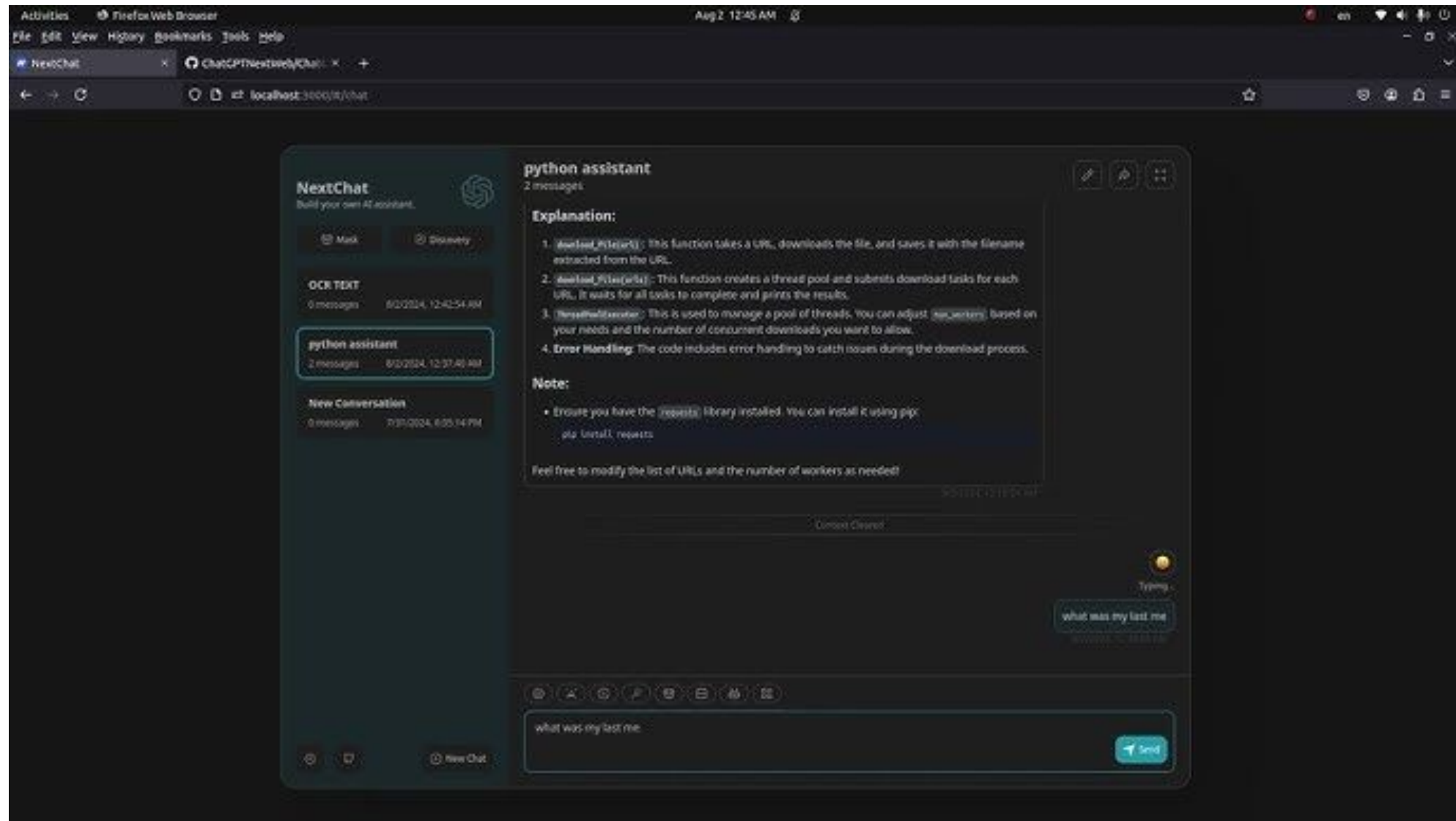
Is the virtual reality / metaverse an enabler for your IoT solution?



And dashboards?



And LLMs?



Systems needed for a company

Data from the devices must go to the usual interfaces from the company and not to a nice / beautiful dashboard.

Where?

- ERP
- MES
- Unified Namespace
- Digital Twin
- LLM
- Excel sheet
- . . .

Security: The CRA (Cyber Resilience Act)

Security: The CRA (Cyber Resilience Act)



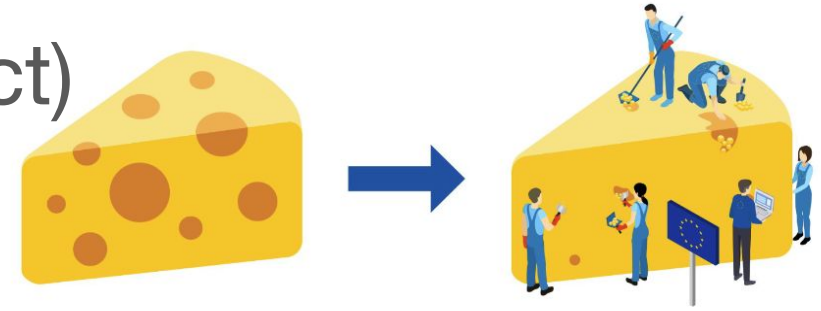
SECURITY

f X in u e

**A single default password
exposes access to dozens of
apartment buildings**

Zack Whittaker — 6:10 AM PST · February 24, 2025

Security: The CRA (Cyber Resilience Act)



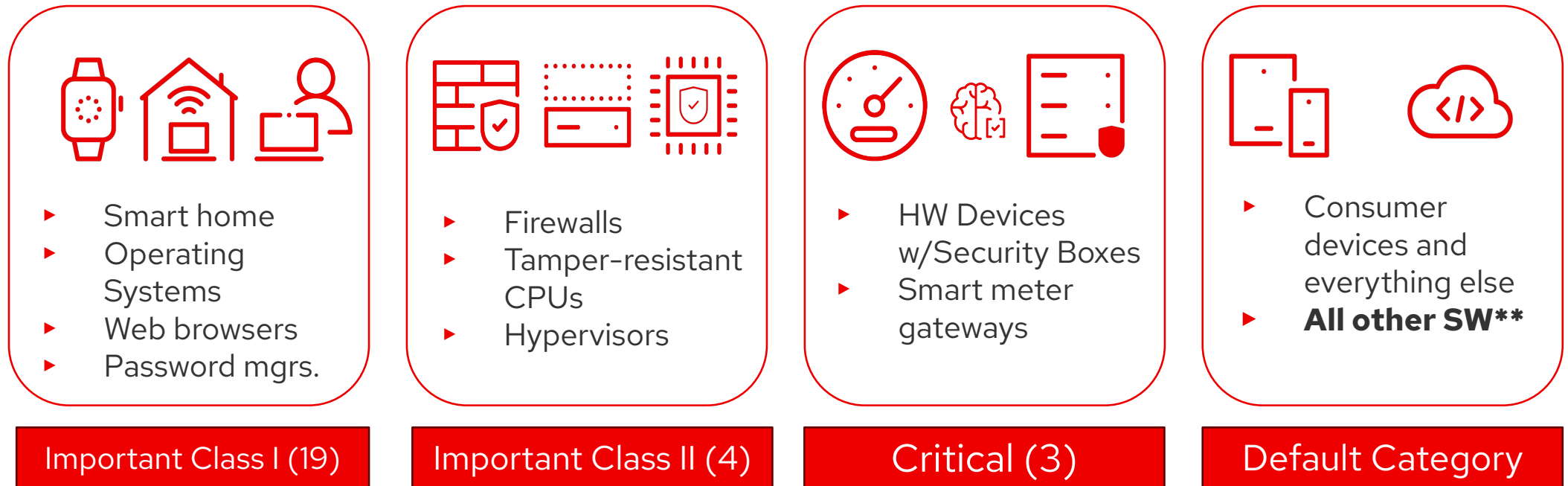
[Cyber Resilience Act \(CRA\)](#) is a EU regulation (applicable worldwide to all who does commercial activity in or with the EU)



- that aims to safeguard consumers and businesses buying or using Products With a Digital Elements (PDE or PwDE - include HW and SW)
- with the introduction of mandatory cybersecurity requirements (mainly) for manufacturers throughout entire PDE lifecycle
- and its supply chain (including 3rd party and open source dependencies).

Security: The CRA (Cyber Resilience Act)

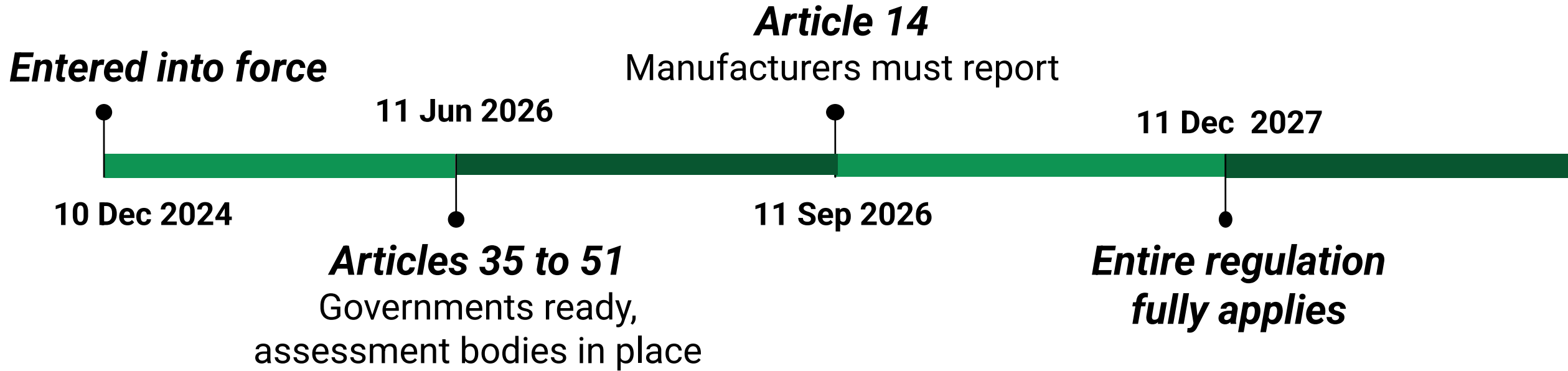
What are the PDEs?



** SW that is PDE as defined in CRA text, placed on the EU market and intended for the course of commercial activity.

The rest 90%

Security: The CRA (Cyber Resilience Act)



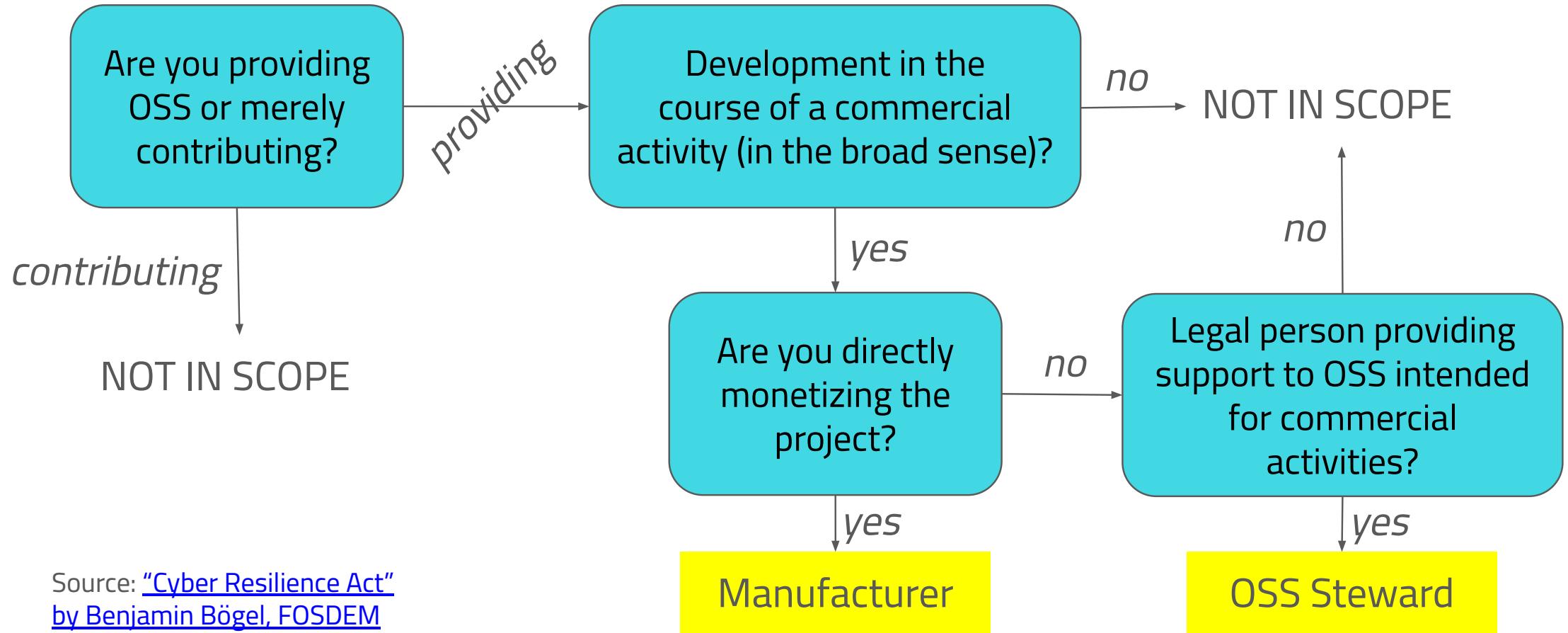
Security: The CRA (Cyber Resilience Act)

Manufacturer's obligations

- Design, develop and produce per essential cybersecurity requirements
- Assess cybersecurity risks, apply throughout lifecycle
- Document/update cybersecurity risk assessment
- Exercise due diligence when integrating 3rd party components including OSS.
- If component vulnerability found, report to component maintainer.
- Various vulnerabilities reporting and handling (usually 5+ years).
- Create technical documentation
- Designate vulnerability single point-of-contact
- Report vulnerabilities and incidents

Security: The CRA (Cyber Resilience Act)

Does the CRA cover your OSS work?



Source: ["Cyber Resilience Act"](#)
by Benjamin Bögel, FOSDEM

Security: The CRA (Cyber Resilience Act)

Reporting obligations

- Manufacturer must notify designated CSIRT and ENISA of any actively exploited vulnerability in PDE that it becomes aware of:
 - Early warning < 24 hours, general info < 72 hours
 - Final report < 14 days after correction/mitigation available
- Must notify for a severe incident:
 - Early warning < 24 hours, incident notification < 72 hours, final report < 1 month
 - Incident is an event compromising data or service. It's severe if it:
 - can negatively affect {availability, authenticity, integrity or confidentiality of} sensitive or important data or functions.
 - can lead to introduction/execution of malicious code in PDE or user network
- Manufacturer must inform impacted users (or all users) after becoming aware of an actively exploited vulnerability or a severe incident “in timely manner”.

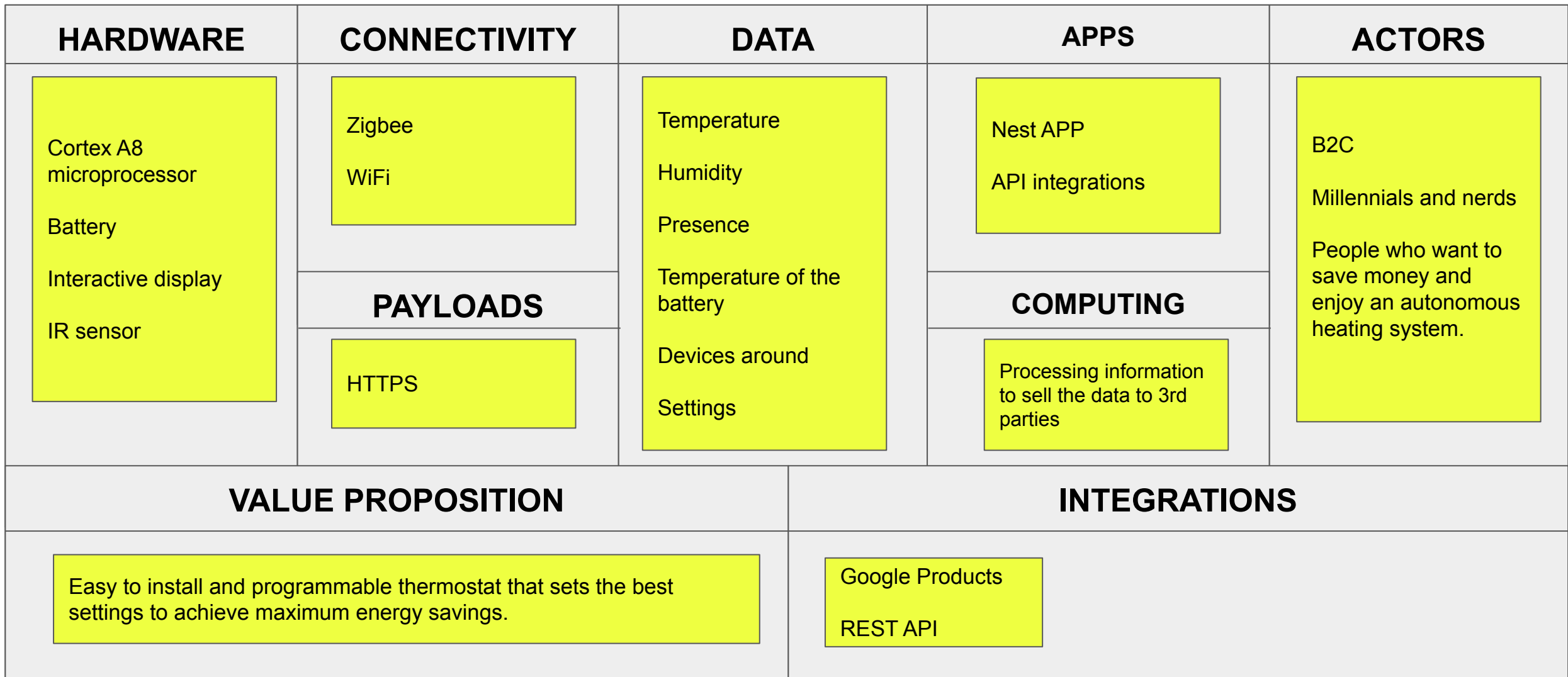
IoT Model Canvas

HARDWARE	CONNECTIVITY	DATA	APPS	ACTORS
	PAYLOADS		COMPUTING	
VALUE PROPOSITION			INTEGRATIONS	

IoT Model Canvas



by Marc Pous [@gy4nt](#)



IoT Model Canvas: Nest Thermostat

HARDWARE	CONNECTIVITY	DATA	APPS	ACTORS
	PAYLOADS		COMPUTING	
VALUE PROPOSITION			INTEGRATIONS	

IoT Model Canvas

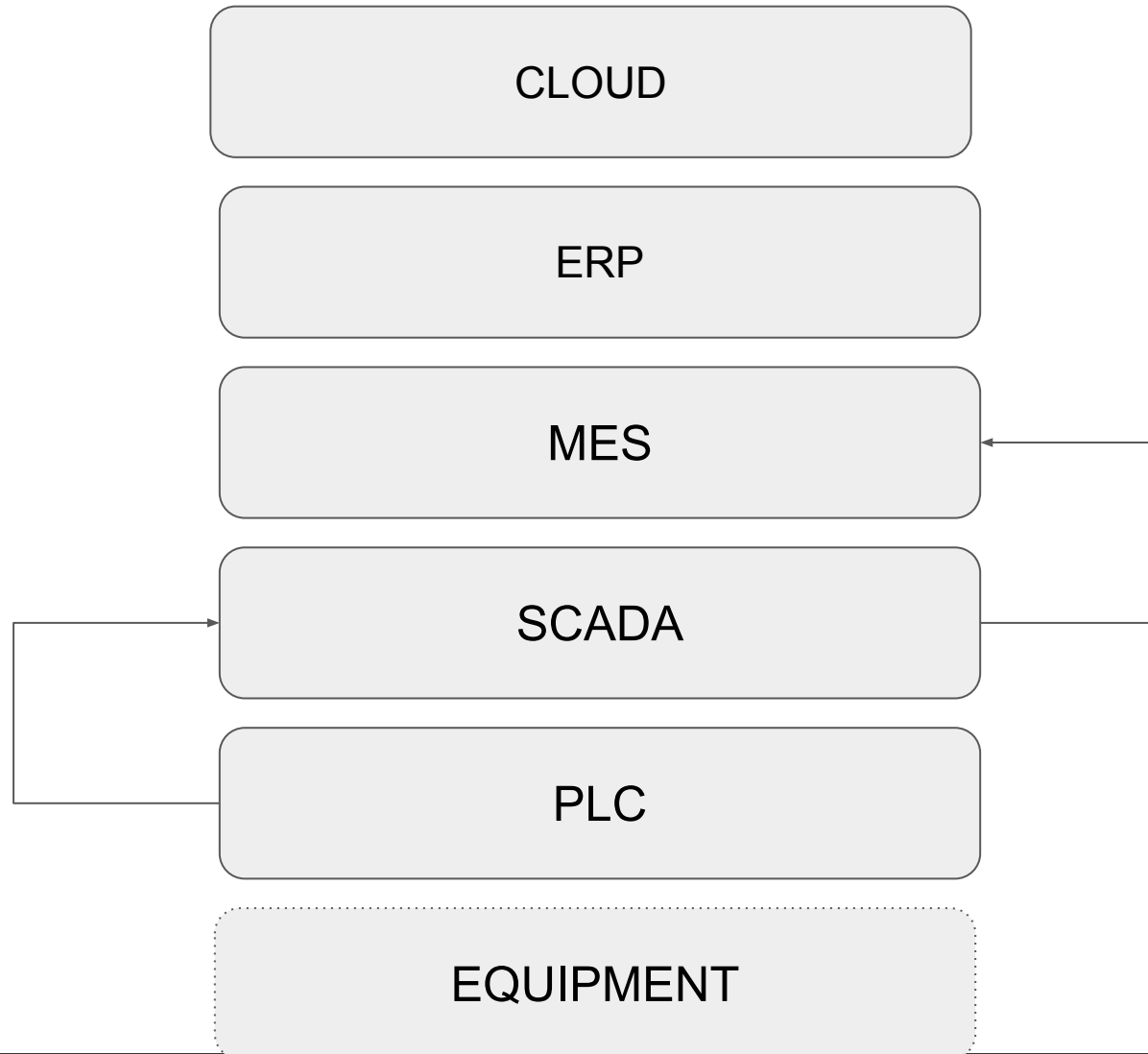


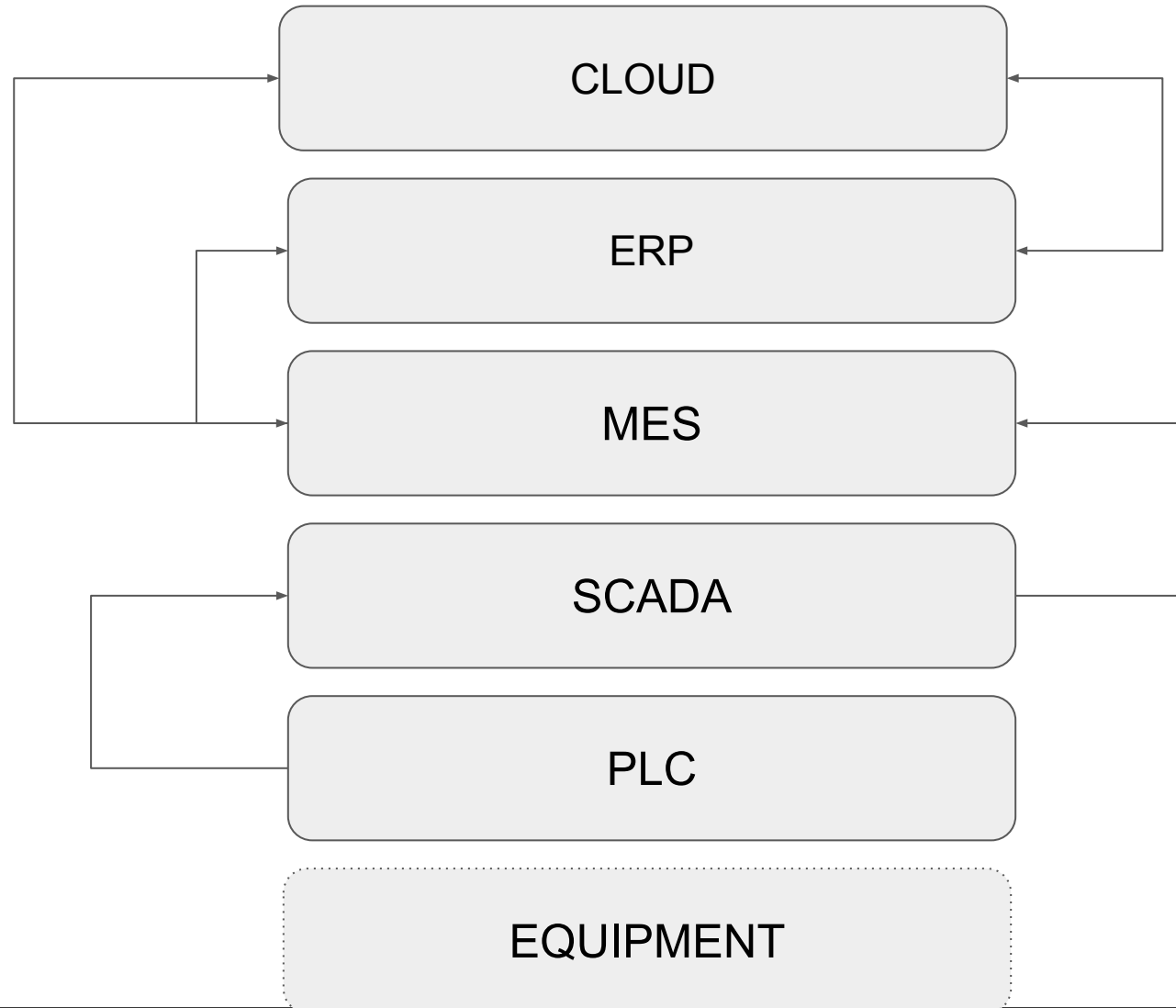
by Marc Pous [@gy4nt](#)

Case: Automation stack on the Industry 4.0

Automation stack on the Industry 4.0







CLOUD

ERP

MES

SCADA

How is this being
integrated with the SCADA?

PLC

PLC₂

EQUIPMENT



How is this being integrated with the SCADA?



OEM



Industry 4.0 goal is to break all of these silos and save money!

How is this being integrated with the SCADA?



OEM

The Unified Namespace (UNS)

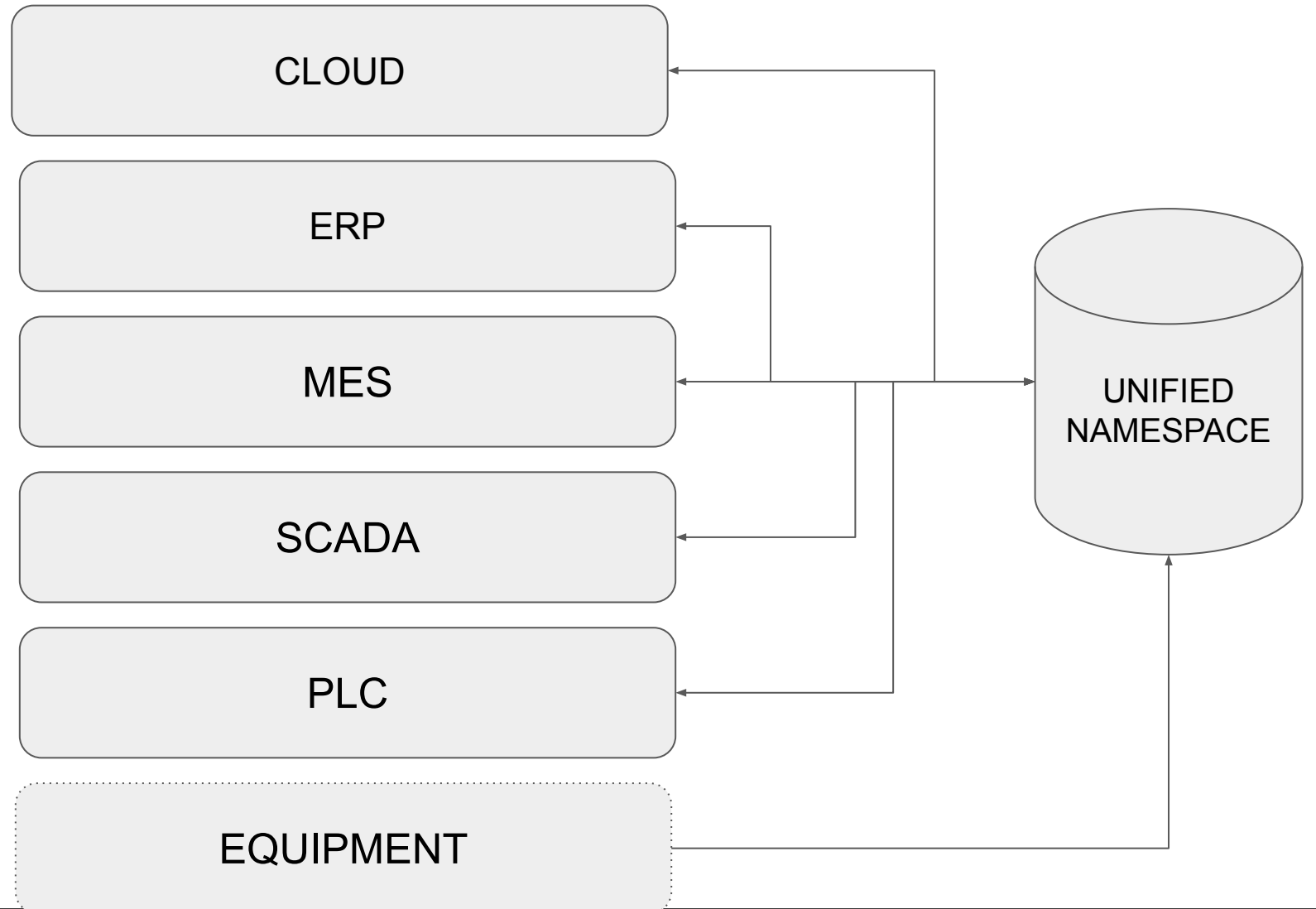
To scale any industrial IoT implementation, the key part is a **unified namespace**.

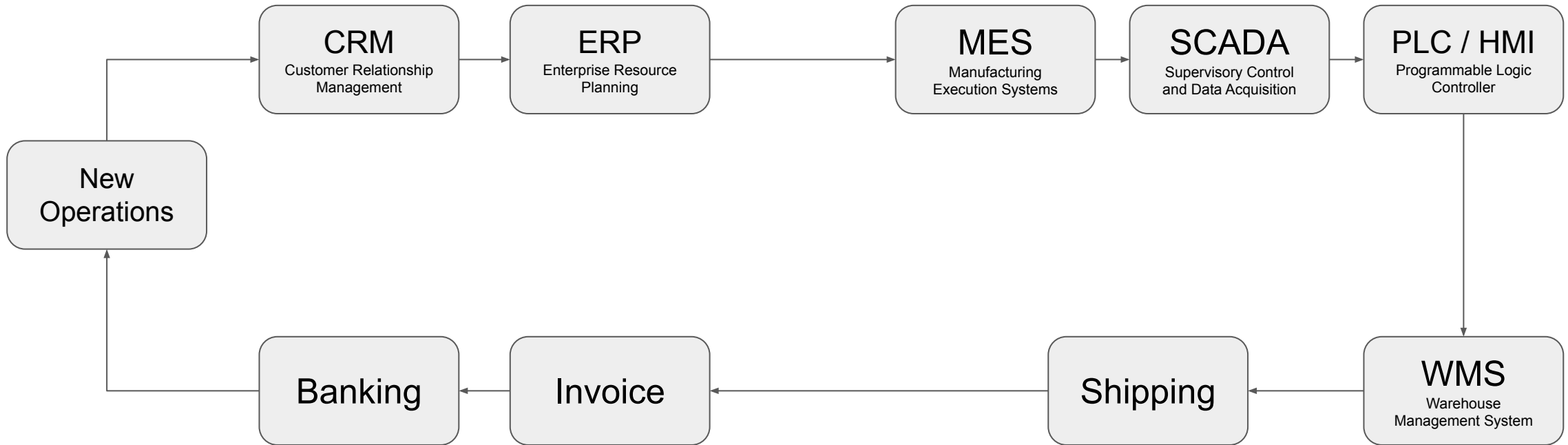
A unified namespace is an abstract layer that acts as a centralized repository of data, information, and context where any application or device can consume or publish data needed for a specific action.

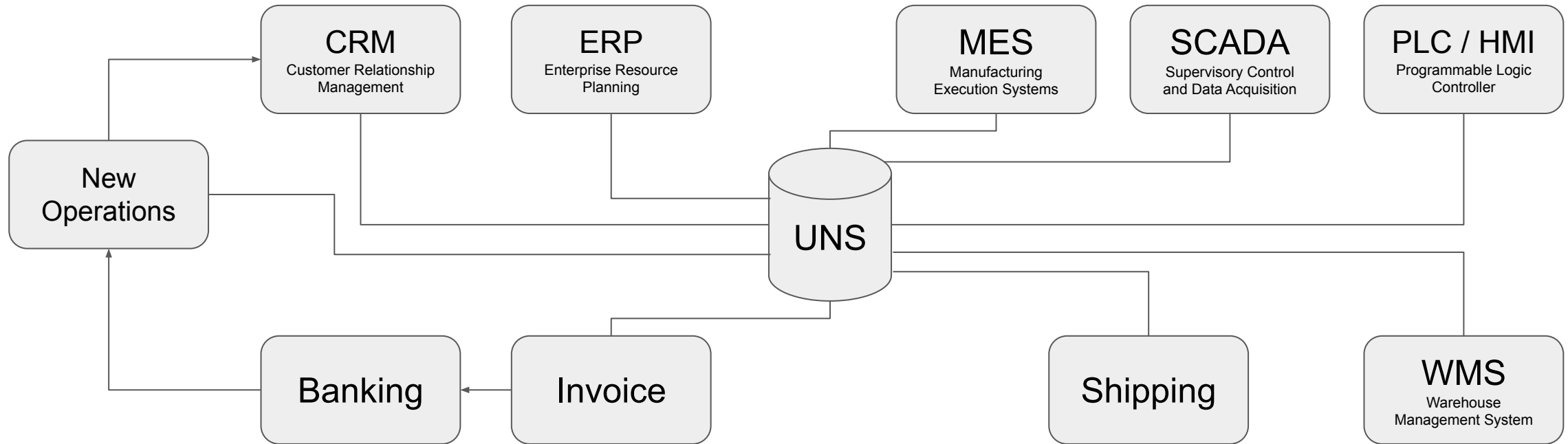
Using a unified namespace everything in the factory acts as a node of the namespace.

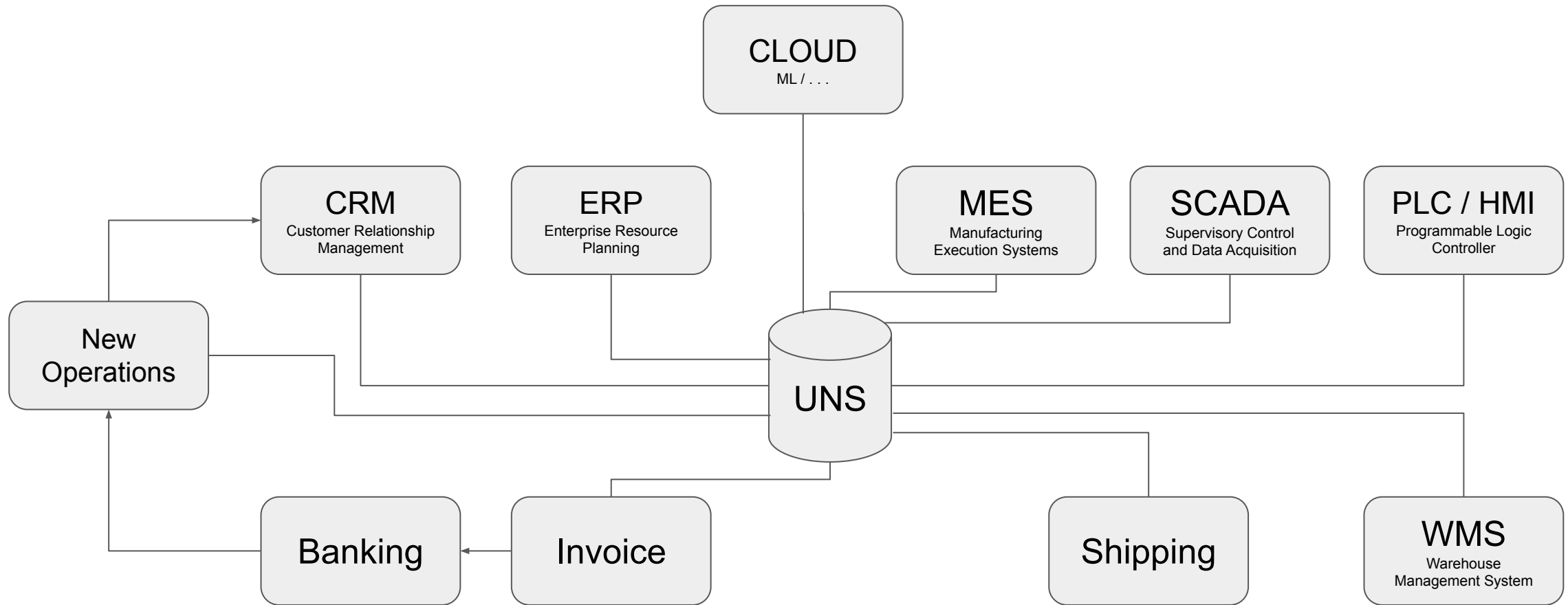
Without a centralized data repository (data warehouse), it could take *months* to deploy a new analytics application across the entire enterprise versus *hours* with a unified namespace.

The Unified Namespace

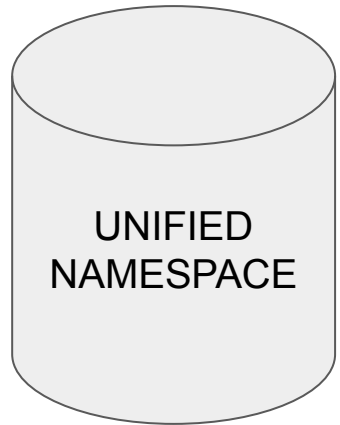








What is a UNS?



On the Industry 4.0, the Unified Namespace is the single source of truth of your enterprise / factory.

Its goal is to bring data from all the proprietary protocols from the factory, in a single way, to a Data Warehouse that manages the IIoT Architecture.

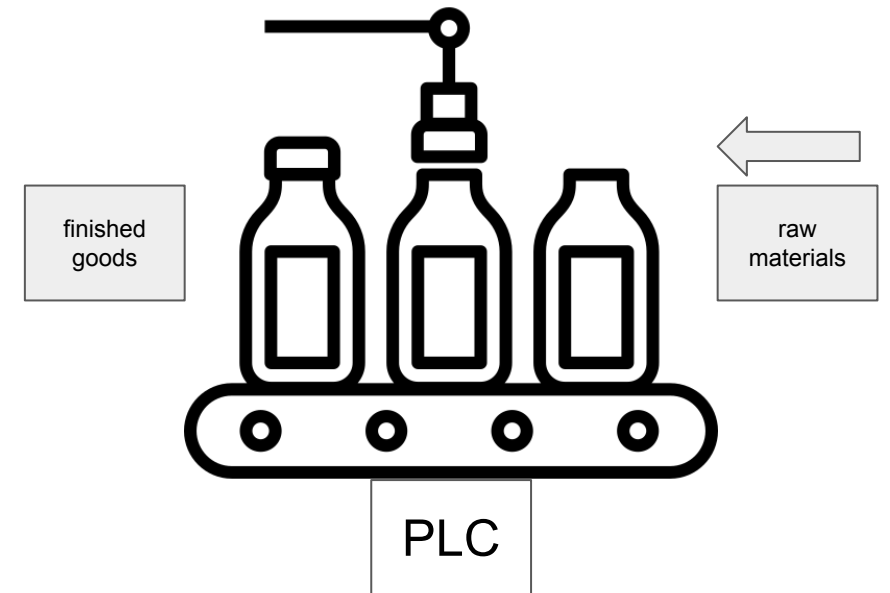
The main technology used in the UNS concept is MQTT.

UNS: The MQTT topics following the ISA-95 standard

- Enterprise
 - Site 1
 - Area 1
 - Line 1
 - Machine 1
 - Machine 2
 - Machine 3
 - Line 2
 - Area 2
 - Area 3
 - Site 2
 - Site 3
 - Site 4
 - Site 5

UNS: The MQTT topics following the ISA-95 standard

- Enterprise
 - Site 1
 - Area 1
 - Line 1
 - Machine 1
 - Machine 2
 - Machine 3
 - Line 2
 - Area 2
 - Area 3
 - Site 2
 - Site 3
 - Site 4
 - Site 5





▼ 127.0.0.1

▶ spBv1.0 (3 topics, 3 messages)

▼ Digital Transformation

Phase = 1

▼ Enterprise

▼ Site 1

▼ Area

▼ Line 1

▼ MES

▼ KPIs

Availability = 0.62

Quality = 0.57

Performance = 0.75

OEE = 0.11

TEEP = 0.01

MTTR = 1381

MTBF = 1418

▼ Edge

Infeed = 48.14

Outfeed = 62.91

Waste = 14.77

State = 1



▼ 127.0.0.1

▶ spBv1.0 (3 topics, 3 messages)

▼ Digital Transformation

Phase = 1

▼ Enterprise

▼ Site 1

▼ Area

▼ Line 1

▼ MES

▼ KPIs

Availability = 0.62

Quality = 0.57

Performance = 0.75

OEE = 0.11

TEEP = 0.01

MTTR = 1381

MTBF = 1418

▼ Edge

Infeed = 48.14

Outfeed = 62.91

Waste = 14.77

State = 1



▼ 127.0.0.1

▶ spBv1.0 (3 topics, 3 messages)

▼ Digital Transformation

Phase = 2

▼ Enterprise

▼ Site 1

▼ Area

▼ Line 1

▶ MES (8 topics, 9 messages)

▶ Edge (4 topics, 4 messages)

▼ ERP

▼ ProductionOrder

OrderNumber = 8826

OrderStatus = "Completed"

ScheduledStartTime = "2023-12-02 19:05:40"

ScheduledEndTime = "2023-12-12 19:05:40"

ActualStartTime = "2023-12-05 19:05:40"

ActualEndTime = "2023-12-06 19:05:40"

OrderedQuantity = 986

ProducedQuantity = 436

RemainingQuantity = 160

▼ Inventory

ItemNumber = 53658

BOM = "data set"

ItemDescription = "Item Description 53658"

AvailableQuantity = 440

ReservedQuantity = 314

OrderedQuantity = 986



▼ 127.0.0.1

▶ spBv1.0 (3 topics, 3 messages)

▼ Digital Transformation

Phase = 3

▼ Enterprise

▼ Site 1

▼ Area

▼ Line 1

▶ MES (19 topics, 20 messages)

▼ Edge

Infeed = 48.14

Outfeed = 62.91

Waste = 14.77

State = 1

▼ Process

SpindleSpeed = 1621

FeedRate = 1.2752394805831337

ToolWear = 0.5070888655308258

CoolantTemperature = 29.181826408532817

Vibration = 1.0772042172363667

PowerConsumption = 46.99291891174136

ToolChangeCount = 4

MaterialTemperature = 66.96252967921029

PartDimensions = "2.0220156547621952x3.9654574643124456"

SurfaceFinish = "Rough"

▼ ERP

▶ ProductionOrder (9 topics, 18 messages)

▶ Inventory (7 topics, 14 messages)



▼ 127.0.0.1

▶ spBv1.0 (3 topics, 3 messages)

▼ Digital Transformation

Phase = 3

▼ Enterprise

▼ Site 1

▼ Area

▼ Line 1

▼ MES

▶ KPIs (8 topics, 9 messages) = 2

▼ Quality

OrderNumber = 1891

ItemNumber = 46718

InspectionResult = "Pass"

RejectionReason = "N/A"

RejectionQuantity = 0

AcceptedQuantity = 74

▼ Maintenance

MachineID = "Machine-68"

MaintenanceStatus = "Overdue"

LastMaintenanceDate = "2023-11-24 19:07:21"

NextMaintenanceDate = "2023-12-21 19:07:21"

MaintenanceHistory = "Last serviced 2023-11-24 19:07:21"

▼ Edge

Infeed = 48.14

Outfeed = 62.91

Waste = 14.77

State = 1



▼ 127.0.0.1

▶ spBv1.0 (3 topics, 3 messages)

▼ Digital Transformation

Phase = 4

▼ Enterprise

▼ Site 1

▼ Area

▼ Line 1

▼ MES

▶ KPIs (8 topics, 9 messages) = 2

▶ Quality (6 topics, 6 messages)

▶ Maintenance (5 topics, 5 messages)

▼ Edge

Infeed = 48.14

Outfeed = 62.91

Waste = 14.77

State = 1

▶ Process (10 topics, 10 messages)

▼ ERP

▶ ProductionOrder (9 topics, 18 messages)

▶ Inventory (7 topics, 14 messages)

55001 = {"AssetLifecycle": {"Status": "Maintenance", "MaintenanceSchedule": "2024-02-11"}, "RiskManagement":

S88 = {"BatchControl": {"EquipmentModule": {"BatchMixingTankStatus": "Mixing", "BottlerStatus": "Maintenance",



MQTT Explorer



- ▼ 127.0.0.1
 - ▶ spBv1.0 (3 topics, 3 messages)
 - ▼ Digital Transformation
 - Phase = 5
 - ▼ Enterprise
 - ▼ Site 1
 - ▼ Area
 - ▼ Line 1
 - ▼ MES
 - ▶ KPIs (8 topics, 9 messages) = 2
 - ▶ Quality (6 topics, 12 messages)
 - ▶ Maintenance (5 topics, 10 messages)
 - ▼ Edge
 - Infeed = 48.14
 - Outfeed = 62.91
 - Waste = 14.77
 - State = 1
 - ▶ Process (10 topics, 20 messages)
 - ▼ ERP
 - ▶ ProductionOrder (9 topics, 18 messages)
 - ▶ Inventory (7 topics, 14 messages)
 - 55001 = {"AssetLifecycle": {"Status": "Decommissioned"}, "BatchControl": {"EquipmentModule": {"BatchMi
 - 588 = {"BatchControl": {"EquipmentModule": {"BatchMi
 - ▶ Line 2 (49 topics, 49 messages)



MQTT Explorer



- ▼ 127.0.0.1
 - ▶ spBv1.0 (3 topics, 3 messages)
 - ▼ Digital Transformation
 - Phase = 6
 - ▼ Enterprise
 - ▼ Site 1
 - ▼ Area
 - ▼ Line 1
 - ▶ MES (19 topics, 31 messages)
 - ▶ Edge (14 topics, 24 messages)
 - ▶ ERP (16 topics, 32 messages)
 - 55001 = {"AssetLifecycle": {"Status": "Decommissioned"}, "BatchControl": {"EquipmentModule": {"BatchMi
 - 588 = {"BatchControl": {"EquipmentModule": {"BatchMi
 - ▼ Line 2
 - ▶ MES (18 topics, 18 messages)
 - ▶ Edge (14 topics, 14 messages)
 - ▶ ERP (15 topics, 15 messages)
 - 55001 = {"AssetLifecycle": {"Status": "Decommissioned"}, "BatchControl": {"EquipmentModule": {"BatchMi
 - 588 = {"BatchControl": {"EquipmentModule": {"BatchMi
 - ▼ Line 3
 - ▶ MES (18 topics, 18 messages)
 - ▶ Edge (14 topics, 14 messages)
 - ▶ ERP (15 topics, 15 messages)
 - 55001 = {"AssetLifecycle": {"Status": "Decommissioned"}, "BatchControl": {"EquipmentModule": {"BatchMi
 - 588 = {"BatchControl": {"EquipmentModule": {"BatchMi

My classes (4)

~~● Business models and the Internet of Things.~~

~~● Business Model Canvas~~

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~~● IoT / Edge AI project (hands on)~~

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~~● The IoT Stack How to build your IoT project, digital transformation and more.~~

~~● IoT Model Canvas~~

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● Scale your IoT / Edge AI projects (hands-on)

EOF